Physics Grade 11 - Solved Exercises (BTBB)

Balochistan Textbook Board (BTBB)

Unit 1: Measurements

Exercise - Multiple Choice Questions (MCQs)

Q1. Which one of the following physical quantities has no unit:

- Torque
- Pressure
- Stress
- ✓ Strain

Q2. The estimated time taken by light to cover the distance 390000 km from moon to Earth with a speed of 3 x 10^8 m s^-1 is:

- 0.77 s
- ✓ 1.3 s
- 5.3 s
- 11.7 s

Q3. The fundamental physical quantities which form the basis of the SI units are:

- Force, weight and time
- ✓ Mass, length and time
- Mass, length and force
- Mass, energy and time

Q4. Which one of the following is base physical quantity?

- Pressure
- ✓ Temperature

Density

Energy

Q5. Light year is the unit of:

Time

✓ Distance

Speed

Velocity

Q6. The derived unit 'joule' in terms of base units is:

kg m s^-2

✓ kg m^2 s^-2

kg m^-1 s^-2

kg m^-2 s^-2

Q7. The length of a body is measured as 3.51 m, if the absolute uncertainty is 0.01 m, then the percentage uncertainty in the measurement is:

3.51 %

0.035 %

✓ 0.28 %

28.65 %

Q8. Which of the following instruments provides most precise measurement:

✓ Screw gauge

Vernier calipers

Measuring tape

Metre ruler

Q9. The dimensions of moment of inertia are:

[MLT^0]

✓ [ML^2T^0]

[ML^-2T^0]

[M^2LT^0]

Q10. Which pair of physical quantities has the same dimensions?

Velocity and acceleration

Mass and weight

Inertia and moment of inertia

✓ Work and potential energy

Exercise - Short Questions

Q1. Describe base and derived units.

Base units are the units of the seven fundamental (base) physical quantities of the SI system, e.g. metre (length), kilogram (mass), second (time), kelvin (temperature), ampere (current), mole (amount of substance) and candela (luminous intensity). Derived units are obtained by combining base units, e.g. the newton (kg m s^-2) for force and the joule (kg m^2 s^-2) for energy.

Q2. Convert your age into seconds.

This is a model answer: a pupil of age 17 years has lived about 17 x 365 days = 6205 days. In seconds this is 6205 x 24 x 3600 = 536,112,000 s, i.e. about 5.36 x 10^8 s. (Each pupil should repeat the same multiplication with his or her own age; leap years add

roughly one extra day per four years.)

Q3. How can you derive the unit of watt in terms of base units?

Power = work / time, so 1 watt = 1 joule per second. Since 1 joule = 1 kg m^2 s^-2, dividing by seconds gives 1 W = 1 kg m^2 s^-2 / s = 1 kg m^2 s^-3. Hence the watt in base units is kg m^2 s^-3.

Q4. Identify three physical quantities which have no units and no dimensions?

Three such dimensionless quantities are: (i) strain (change in length / original length), (ii) refractive index (speed of light in vacuum / speed of light in the medium), and (iii) relative density (density of a substance / density of water). In each case the units cancel in the ratio, so the quantity is pure and dimensionless.

Q5. Which one is more reliable between precise and accurate measurement?

An accurate measurement is more reliable. Precision only means that repeated readings agree closely with each other; they may all be wrong together (e.g. a wrongly zeroed instrument). Accuracy means closeness to the true value, which is what a reliable result needs. The best measurement is both precise and accurate.

Q6. How can you minimize the uncertainty in the reported result?

Uncertainty can be minimized by: (i) repeating the measurement several times and taking the mean value, (ii) using an instrument with a smaller least count, (iii) removing zero error and parallax error by careful technique, and (iv) keeping the experimental

conditions steady. Averaging repeated readings cancels random errors and narrows the range of uncertainty.

Q7. How can accuracy be increased by decreasing the limit of precision?

Decreasing the limit of precision means using an instrument of smaller least count, i.e. a finer scale. A finer scale reads smaller differences, so the fractional uncertainty (absolute uncertainty / measured value) falls and the result lies closer to the true value; that is, the accuracy of the measurement increases.

Q8. Write down three different physical quantities who have the same dimensions?

Work, kinetic energy and potential energy all have the same dimensions [ML^2T^-2] and the same unit, the joule. Work = force x distance, kinetic energy = (1/2) m v^2 and gravitational potential energy = m g h all reduce to mass x length^2 / time^2.

Exercise - Comprehensive Questions

Q1. Define estimation of physical quantities? Discuss the estimation of base physical quantities.

Estimation means finding the approximate value (order of magnitude) of a quantity without an exact measurement, using known reference values and simple reasoning. Base quantities are estimated from familiar standards: length by comparing with a metre rule or the known height of a person, mass by comparing with standard masses, and time by counting a regular event (e.g. a pendulum swing or heartbeat). Such estimates are rounded to one

significant figure and are useful for checking whether a calculated result is reasonable.

Q2. Describe base and derived units.

Base units are the seven independent SI units: metre (m) for length, kilogram (kg) for mass, second (s) for time, ampere (A) for current, kelvin (K) for temperature, mole (mol) for amount of substance and candela (cd) for luminous intensity. All other units are derived from them by multiplication or division: e.g. speed (m s^-1), force = mass x acceleration = kg m s^-2 (newton), and energy = force x distance = kg m^2 s^-2 (joule). Derived units may carry special names, but every one can be written in base units.

Q3. Differentiate between precision and accuracy with examples.

Precision is the closeness of repeated measurements to one another; accuracy is the closeness of a measurement to the true value. Example: four darts all landing together but far from the bull's-eye are precise but not accurate; four darts spread around the bull's-eye are accurate on average but not precise; four darts clustered on the bull's-eye are both precise and accurate. Precision depends on the least count of the instrument, while accuracy depends on freedom from systematic error.

Q4. What do you know about the dimensions of physical quantities? How can you explain the nature of physical quantities by using the dimensional analysis?

The dimensions of a quantity show how it is built from the base

quantities mass M, length L and time T (and others where needed), e.g. velocity [LT^-1], force [MLT^-2] and energy [ML^2T^-2]. Dimensional analysis reveals the nature of a quantity: two quantities with the same dimensions are of the same kind (e.g. work and energy), a quantity whose dimensions contain no M, L or T is dimensionless (e.g. strain), and an equation can only be correct if every term has the same dimensions (principle of homogeneity).

Q5. What is the role of dimensional analysis in the derivation of a formula and homogeneity of a mathematical equation?
Dimensional analysis derives a formula by assuming the unknown quantity depends on certain variables raised to unknown powers, equating dimensions on both sides, and solving for the powers (e.g. deriving F = m v^2 / r for centripetal force). It also tests homogeneity: in any correct equation all terms added or equated must have identical dimensions, so a dimensionally inhomogeneous equation is certainly wrong. It cannot fix pure numbers (like 2 pi) or distinguish quantities of the same dimensions.

Exercise - Numerical Problems
Q1. It has been estimated that 150 workers could dug a tunnel through a mountain in 25 days. Estimate that how long would be completed this project by 250 workers. (Book answer: 15 days)
Total work = number of workers x number of days = 150 x 25 = 3750 worker-days.

With 250 workers, time = total work / workers = 3750 / 250.

3750 / 250 = 15 days.

The tunnel will be completed in 15 days. This matches the book answer.

Q2. The length of a second pendulum is 0.99 m with its time period 2 seconds. When its length is reduced to half then estimate its time period and frequency. (Book answer: 1.4 s, 0.71 Hz)

For a simple pendulum T = 2 pi sqrt(L / g), so T is proportional to sqrt(L).

New length L' = 0.99 / 2 = 0.495 m, i.e. half the old length.

T' = T / sqrt(2) = 2 / 1.414 = 1.41 s, about 1.4 s.

Frequency f' = 1 / T' = 1 / 1.41 = 0.71 Hz.

Time period about 1.4 s and frequency 0.71 Hz. This matches the book answer.

Q3. How much distance is covered by light in one year when its speed in space is 3 x 10^8 m s^-1? (Book answer: 9.5 x 10^15 m)

Time in one year = 365 x 24 x 3600 = 31,536,000 s = 3.1536 x 10^7 s.

Distance = speed x time = 3 x 10^8 x 3.1536 x 10^7.

3 x 3.1536 = 9.4608, and 10^8 x 10^7 = 10^15.

Distance = 9.46 x 10^15 m, about 9.5 x 10^15 m (one light year).

About 9.5 x 10^15 m. This matches the book answer.

Q4. Prove that nano seconds in one second is more than the number of seconds in one year.

Number of nanoseconds in one second = 1 s / 10^-9 s = 10^9 = 1,000,000,000.

Number of seconds in one year = 365 x 24 x 3600 = 31,536,000 = 3.1536 x 10^7.

Compare: 10^9 > 3.1536 x 10^7, since 1,000,000,000 > 31,536,000.

Hence the nanoseconds in one second (10^9) are far more than the seconds in one year (about 3.15 x 10^7). Proved.

Q5. Convert the following: (a) 20 m s^-1 into km h^-1, (b) 3 x 10^8 m s^-1 into km h^-1, (c) 220 km h^-1 into m s^-1. (Book answers: (a) 72 km h^-1, (b) 1.08 x 10^10 km h^-1, (c) 61 m s^-1)

Conversion factor: 1 m s^-1 = 3.6 km h^-1, and 1 km h^-1 = 1 / 3.6 = 5 / 18 m s^-1.

(a) 20 m s^-1 = 20 x 3.6 = 72 km h^-1.

(b) 3 x 10^8 m s^-1 = 3 x 10^8 x 3.6 = 10.8 x 10^8 = 1.08 x

10^9 km h^-1.

(c) 220 km h^-1 = 220 x 5 / 18 = 1100 / 18 = 61.1 m s^-1, about 61 m s^-1.

Answers: (a) 72 km h^-1, (b) 1.08 x 10^9 km h^-1, (c) about 61 m s^-1. NOTE: the book prints 1.08 x 10^10 km h^-1 for part (b), which is ten times too large; the correct value 1.08 x 10^9 km h^-1 is obtained above and verified by 3 x 10^8 x 3600 = 1.08 x 10^12 m h^-1 = 1.08 x 10^9 km h^-1.

Q6. Verify that the equation S = v_i t + a t^2 is dimensionally correct. (The standard form carries (1/2) a t^2; the factor 1/2 is dimensionless and does not affect dimensions.)
Dimensions of S (displacement) = [L].

Dimensions of v_i t = [LT^-1] x [T] = [L].

Dimensions of a t^2 = [LT^-2] x [T^2] = [L].

Every term has dimensions [L], so both sides agree.

The equation is dimensionally homogeneous, hence dimensionally correct.

Q7. The centripetal force 'F_c' is acting on a particle of mass 'm' which is moving with velocity 'v' along a circle of radius 'r', where the applied force depends on the mass 'm' of the particle, its

velocity 'v' and radius 'r' of the circle. Derive dimensionally the formula for the centripetal force 'F_c' by using dimension analysis.
Assume F_c = k m^a v^b r^c, where k is a dimensionless constant.

Dimensions: [MLT^-2] = [M]^a [LT^-1]^b [L]^c = M^a L^(b+c) T^(-b).

Equate powers: for M: a = 1; for T: -b = -2, so b = 2; for L: b + c = 1, so c = 1 - 2 = -1.

Thus F_c = k m^1 v^2 r^-1 = k m v^2 / r. Experiment gives k = 1.

F_c = m v^2 / r.

Q8. Find the dimensions of gravitational constant G by using the relation, F = G m1 m2 / r^2.
Rearrange: G = F r^2 / (m1 m2).

Dimensions of F = [MLT^-2]; of r^2 = [L^2]; of m1 m2 = [M^2].

Dimensions of G = [MLT^-2] x [L^2] / [M^2] = M^(1-2) L^(1+2) T^-2.

G = [M^-1 L^3 T^-2].

The dimensions of G are [M^-1 L^3 T^-2].

Q9. According to Hook's law, the restoring force F due to a body attached to a spring is given by F = -k x. Calculate the dimensions of the spring constant k.

From F = -k x, the magnitude gives k = F / x (the minus sign only gives direction).

Dimensions of F = [MLT^-2]; dimensions of x (extension) = [L].

Dimensions of k = [MLT^-2] / [L] = [MT^-2].

The spring constant has dimensions [MT^-2] and unit N m^-1.

Unit 2: Vectors and Scalar Quantities

Exercise - Multiple Choice Questions (MCQs)

Q1. Which pair includes a scalar and a vector quantity?

- ✓ K.E. and momentum
- P.E. and Work
- Weight and force
- Velocity and acceleration

Q2. The magnitudes of two vectors are 3 N and 4 N respectively. If the angle between them is 90 degrees, then the magnitude of their resultant vector will be:

- ✓ 5 N
- 6 N
- 7 N
- Zero

Q3. At what angle the vertical component of a vector is maximum?

- 0 degrees
- 30 degrees
- 45 degrees
- ✓ 90 degrees

Q4. In which quadrant, a vector will be drawn when its both x and y components are negative.

- First

Second

✓ Third

Fourth

Q5. What is the possible answer of (-3 i-hat) . (-4 j-hat)?

✓ Zero

One

12 k-hat

12

Q6. The angle of the given vector 2 i-hat + 2 j-hat is:

30 degrees

✓ 45 degrees

60 degrees

90 degrees

Q7. The correct answer for the expression j-hat . (k-hat x i-hat) is:

Zero

✓ One

i-hat

j-hat

Q8. The scalar product of two non-zero vectors is equal to zero when angle theta between them is:

0 degrees

30 degrees

60 degrees

✓ 90 degrees

Q9. The magnitude of vector product of two vectors parallel to each other is:

✓ 0

0.5

+1

-1

Q10. The vector product of two vectors is maximum when both the vectors are:

Parallel

Anti-parallel

✓ Perpendicular

Equal

Q11. What is the expected result of (A vector x B vector)^2 + (A vector . B vector)^2 = ?

A^2 B^2 cos theta

A^2 B^2 sin theta

AB

✓ A^2 B^2

Exercise - Short Questions

Q1. Why a scalar quantity cannot be added or subtracted with a vector quantity?

A scalar has magnitude only while a vector has magnitude and direction. Adding them would mix two different kinds of quantity: the result could not be assigned a single direction, nor a pure

magnitude, and the operation would violate the principle of homogeneity. Only like quantities (scalar with scalar, vector with vector) can be added or subtracted.

Q2. Why the division of a vector by another vector is impossible?

Division by a vector is impossible because a vector has no unique inverse: many different vectors could be claimed as the quotient, since direction cannot be divided out. There is no vector V such that dividing by it undoes multiplication in a single well-defined way, so the operation is not defined in vector algebra.

Q3. Under what conditions, the rectangular components of a vector have the same magnitude?

The rectangular components are A_x = A cos theta and A_y = A sin theta. They are equal when cos theta = sin theta, i.e. when theta = 45 degrees (in the first quadrant). Then A_x = A_y = A / sqrt(2).

Q4. Can the scalar product of two vector quantities be negative? If your answer is yes, give an example, if no give a reason?

Yes. The scalar product A . B = AB cos theta is negative when cos theta is negative, i.e. when the angle between the vectors is between 90 degrees and 270 degrees. Example: the work done by friction is negative, because the friction force acts opposite to the displacement (theta = 180 degrees, cos theta = -1).

Q5. How does a scalar product of two vectors obey the commutative law?

The scalar product obeys the commutative law because A . B = AB

cos theta and B . A = BA cos theta, and AB = BA and the angle between them is the same whichever order is taken. Hence A . B = B . A always.

Q6. How can you determine the direction of the resultant vector due to the vector product of two vectors?

The direction of the vector product C = A x B is found by the right-hand rule: point the fingers of the right hand along A and curl them towards B; the extended thumb points along C. Equivalently, C is perpendicular to the plane containing A and B, in the direction in which a right-handed screw advances when turned from A to B.

Exercise - Comprehensive Questions

Q1. Differentiate between the scalar and vector physical quantities? Explain the representation of a vector quantity.

Scalar quantities have magnitude only: mass, time, temperature, energy and speed. Vector quantities have magnitude and direction: displacement, velocity, force and momentum. A vector is represented graphically by a straight line with an arrowhead: the length of the line (to a chosen scale) gives the magnitude and the arrowhead gives the direction. It is also written in bold (A) or with an arrow above it, and in unit-vector form as A = A_x i-hat + A_y j-hat + A_z k-hat.

Q2. How can you resolve a vector into its perpendicular components? Explain.

A vector A making an angle theta with the x-axis is resolved into two perpendicular components: A_x = A cos theta along the x-axis

and A_y = A sin theta along the y-axis. Geometrically this is done by dropping a perpendicular from the tip of the vector onto each axis; the two projections are the components. The original vector is recovered as A = sqrt(A_x^2 + A_y^2) with direction theta = tan^-1 (A_y / A_x). The components act together exactly like the original vector.

Q3. State and explain the scalar product of two vectors with its properties.

The scalar (dot) product of two vectors is A . B = AB cos theta, a scalar quantity. Its properties are: (i) it is commutative, A . B = B . A; (ii) it is distributive over addition, A . (B + C) = A . B + A . C; (iii) it is zero for perpendicular vectors (cos 90 degrees = 0) and maximum (AB) for parallel vectors; (iv) the dot product of a vector with itself gives the square of its magnitude, A . A = A^2; and (v) in unit-vector form, A . B = A_x B_x + A_y B_y + A_z B_z, since i-hat . i-hat = 1 and i-hat . j-hat = 0, etc.

Q4. Explain in detail the vector product of two vectors. Discuss the properties of vector product.

The vector (cross) product of two vectors is C = A x B = AB sin theta n-hat, where n-hat is the unit vector perpendicular to the plane of A and B given by the right-hand rule; it is a vector quantity. Its properties are: (i) it is anti-commutative, A x B = - (B x A); (ii) it is distributive over addition; (iii) it is zero for parallel vectors (sin 0 degrees = 0) and maximum (AB) for perpendicular vectors; (iv) A x A = 0; (v) its magnitude equals the area of the parallelogram formed by A and B; and (vi) in unit-vector form it is given by the

determinant with rows (i-hat, j-hat, k-hat), (A_x, A_y, A_z), (B_x, B_y, B_z).

Exercise - Numerical Problems

Q1. Find the projection of vector A = 2 i-hat + 4 j-hat on the vector B = 4 i-hat + 3 k-hat. (Book answer: 8/5)

Projection of A on B = (A . B) / |B|.

A . B = (2)(4) + (4)(0) + (0)(3) = 8.

|B| = sqrt(4^2 + 0^2 + 3^2) = sqrt(25) = 5.

Projection = 8 / 5.

The projection is 8/5 units. This matches the book answer.

Q2. Find the x and y components of a vector A = 4 i-hat + 7 j-hat and it makes an angle 60 degrees with x-axis. (Book answer: 4, 7)

A vector written as A_x i-hat + A_y j-hat has x-component A_x and y-component A_y directly.

Here A_x = 4 and A_y = 7.

Check: tan theta = A_y / A_x = 7/4 = 1.75, so theta = tan^-1(1.75) = 60.25 degrees, about the printed 60 degrees.

The x and y components are 4 and 7. This matches the book answer. NOTE: the printed angle of 60 degrees is only approximate

for this vector: tan theta = 7/4 gives theta = 60.25 degrees, about 60 degrees.

Q3. Prove that the three vectors 3 i-hat + j-hat + 2 k-hat, i-hat - j-hat - k-hat and i-hat + 5 j-hat - 4 k-hat are at right angle to one another.

Two vectors are at right angles if their scalar product is zero.

First pair: (3)(1) + (1)(-1) + (2)(-1) = 3 - 1 - 2 = 0.

Second pair: (3)(1) + (1)(5) + (2)(-4) = 3 + 5 - 8 = 0.

Third pair: (1)(1) + (-1)(5) + (-1)(-4) = 1 - 5 + 4 = 0.

All three dot products are zero, so each pair is perpendicular.

The three vectors are mutually perpendicular, i.e. each is at right angle to the other two. Proved.

Q4. The magnitude of dot and cross products of two vectors are 6 and 6 sqrt(3) respectively. Find the angle between them. (Book answer: 60 degrees)

A . B = AB cos theta = 6, and |A x B| = AB sin theta = 6 sqrt(3).

Divide: (AB sin theta) / (AB cos theta) = tan theta = 6 sqrt(3) / 6 = sqrt(3).

theta = tan^-1(sqrt(3)) = 60 degrees.

The angle between them is 60 degrees. This matches the book answer.

Q5. A force, F = 4 i-hat - 3 j-hat + 2 k-hat acts on an object and the object is displaced along a straight line from point A(3, 2, -1) to point B(2, -1, 4). If force is measured in newton and displacement in metres then find work done on the object. (Book answer: 32 J)
Displacement d = B - A = (2 - 3) i-hat + (-1 - 2) j-hat + (4 - (-1)) k-hat = -i-hat - 3 j-hat + 5 k-hat.

Work W = F . d = (4)(-1) + (-3)(-3) + (2)(5).

(4)(-1) = -4; (-3)(-3) = +9; (2)(5) = +10.

W = -4 + 9 + 10 = 15 J.

The work done is 15 J. NOTE: the book prints 32 J, which does not follow from the printed data; the step-by-step dot product below gives 15 J, verified by recomputation.

Unit 3: Mechanics

Exercise - Multiple Choice Questions (MCQs)

Q1. Rate of change of momentum is equal to:

- Impulse
- ✓ Force
- Torque
- Velocity

Q2. At what angle a projectile gains its maximum height?

- 0 degrees
- 45 degrees
- 60 degrees
- ✓ 90 degrees

Q3. A rocket works on basis of:

- Newton's 1st law
- Newton's 2nd law
- ✓ Newton's 3rd law
- Newton's gravitational law

Q4. The vertical height and horizontal range of the projectile will be equal if angle of projection is:

- 45 degrees
- 56 degrees
- 66 degrees

✓ 76 degrees

Q5. A man is pulling a bag with force of 15 N at an angle 60 degrees with horizontal plane. If bag covers a distance of 10 m, then work done by the man is:

50 J

✓ 75 J

100 J

150 J

Q6. The area under a curved line in a force and displacement graph shows that the work done under a:

constant force

✓ variable force

maximum force

minimum force

Q7. If the speed of an object is tripled, its kinetic energy is increased by:

1/9 times

1/3 times

6 times

✓ 9 times

Q8. When the K.E of a body is decreased by an applied force, its corresponding work is taken as:

Positive

✓ Negative

Constant

Zero

Q9. A body of mass 100 g is raised vertically from surface of Earth in a gravitational field. The P.E. of the body at height 100 m is:

0.98 J

9.8 J

✓ 98 J

980 J

Q10. Which one of the following forces is a non-conservative force?

Gravitational force

✓ Friction force

Electrostatic force

Magnetic force

Exercise - Short Questions

Q1. State the law of conservation of linear momentum.

The law of conservation of linear momentum states: in the absence of an external force, the total momentum of a system of interacting bodies remains constant. That is, the total momentum before a collision equals the total momentum after it: m1 u1 + m2 u2 = m1 v1 + m2 v2.

Q2. How is elastic collision different from inelastic collision?

In an elastic collision both momentum and kinetic energy are conserved, and the bodies separate with no loss of energy. In an inelastic collision momentum is conserved but kinetic energy is not:

part of it is converted into heat, sound or deformation, and the bodies may stick together (perfectly inelastic collision) or separate with reduced speed.

Q3. When the velocity of the projectile is minimum and maximum on its trajectory?

The speed of a projectile is maximum at the points of projection and landing, where it equals the initial speed v_i, and minimum at the highest point of the trajectory, where the vertical component of velocity is zero and only the horizontal component v_i cos theta remains.

Q4. How can a body achieve its maximum range in a projectile motion?

The horizontal range is R = v_i^2 sin 2 theta / g, which is maximum when sin 2 theta = 1, i.e. 2 theta = 90 degrees, so theta = 45 degrees. A body achieves maximum range by being projected at 45 degrees to the horizontal (in the absence of air resistance).

Q5. State the values of projected angle at which the horizontal range of the two trajectory paths is same.

The range is the same for complementary angles: theta and (90 degrees - theta) give the same range. Examples: 30 degrees and 60 degrees; 15 degrees and 75 degrees; 45 degrees with any pair (45 degrees + theta) and (45 degrees - theta). In general, any two angles adding to 90 degrees give equal ranges.

Q6. Calculate the work done by a body when it is moving with

uniform velocity.

For a body moving with uniform velocity there is no change in velocity, so acceleration is zero and the net force is zero. Hence the work done by the net force is W = F s cos theta = 0. (Individual forces may do work that cancels out, but the net work is zero, consistent with no change in kinetic energy.)

Q7. Why the value of K.E. is always positive?

Kinetic energy is K.E. = (1/2) m v^2. Mass is always positive and v^2 is always positive (the square of a speed, positive or negative velocity alike), so their product can never be negative. Kinetic energy is therefore always positive (or zero at rest).

Q8. What are the three properties of a conservative force?

The three properties of a conservative force are: (i) the work done by it between two points is independent of the path taken; (ii) the work done by it around any closed path is zero; and (iii) it can be expressed as the negative gradient of a potential energy, so that mechanical energy (K.E. + P.E.) is conserved under it. Examples: gravitational and electrostatic forces.

Q9. A bucket is taken to the bottom of a well, does the bucket possess any potential energy.

Yes, it still possesses potential energy, but measured from a new reference level: potential energy is always relative. Taking the bottom of the well as the reference, the bucket there has zero potential energy relative to that level, but relative to the ground surface it has negative potential energy (-m g h, where h is the

depth of the well). What matters physically is the change in potential energy between two positions.

Exercise - Comprehensive Questions

Q1. Derive analytically and graphically the three equations of motion using velocity-time graph.

Analytically, with constant acceleration a: from a = (v_f - v_i)/t comes v_f = v_i + a t (first equation). The average velocity is (v_i + v_f)/2, so displacement S = ((v_i + v_f)/2) t; substituting v_f gives S = v_i t + (1/2) a t^2 (second equation). Eliminating t between the two gives v_f^2 = v_i^2 + 2 a S (third equation). Graphically, on a velocity-time graph, the slope of the line is acceleration (giving the first equation), the area under the line is displacement: the rectangle v_i t plus the triangle (1/2) (v_f - v_i) t gives S = v_i t + (1/2) a t^2, and combining these gives the third equation.

Q2. What is elastic and inelastic collisions in one dimension? Calculate the velocities of the bodies after their collision and discuss these final velocities under different cases.

In a one-dimensional elastic collision both momentum and kinetic energy are conserved. For masses m1, m2 with initial velocities u1, u2, the final velocities are v1 = ((m1 - m2) u1 + 2 m2 u2)/(m1 + m2) and v2 = ((m2 - m1) u2 + 2 m1 u1)/(m1 + m2). In an inelastic collision only momentum is conserved; kinetic energy is partly lost. Special cases: (i) equal masses, one at rest: the moving body stops and the other moves off with its velocity; (ii) a light body hitting a heavy body at rest rebounds with nearly its own speed; (iii) a heavy body hitting a light body at rest carries it forward at nearly twice its

own speed. In a perfectly inelastic collision the bodies stick together and move with the common velocity V = (m1 u1 + m2 u2)/(m1 + m2).

Q3. State and explain elastic and inelastic collisions in two dimensions.

In a two-dimensional elastic collision the bodies scatter in different directions in a plane. Momentum is conserved separately along each axis (x and y), and kinetic energy is conserved. In a two-dimensional inelastic collision momentum is still conserved along each axis, but kinetic energy is not conserved: part of it becomes heat, sound or deformation. If the bodies stick together they move off as one body with the vector velocity given by total momentum divided by total mass.

Q4. What is projectile and projectile motion? Discuss displacement, velocity and acceleration of the projectile along its trajectory path.

A projectile is a body thrown into the air which then moves under gravity alone (air resistance neglected). Projectile motion is the resulting two-dimensional motion. Along the trajectory: the horizontal displacement grows uniformly, x = v_i cos theta x t, because there is no horizontal acceleration; the vertical displacement is y = v_i sin theta x t - (1/2) g t^2, under constant downward acceleration g; the horizontal velocity stays constant at v_i cos theta while the vertical velocity decreases as v_y = v_i sin theta - g t, becoming zero at the top and then negative; the acceleration is g vertically downward at every point of the path.

Q5. Explain the characteristics of projectile motion such as time of flight, maximum height and horizontal range.

The characteristics of projectile motion are: (i) Time of flight T = 2 v_i sin theta / g, the time from projection to return to the same level; (ii) Maximum height H = v_i^2 sin^2 theta / (2 g), reached when the vertical velocity becomes zero; (iii) Horizontal range R = v_i^2 sin 2 theta / g, maximum at theta = 45 degrees and equal for complementary angles. All three increase with the initial speed; the height and time increase with sin theta while the range is symmetric about 45 degrees.

Q6. Define work done by a constant force and explain with examples, positive work, negative work and zero work.

Work done by a constant force is W = F . S = F S cos theta, the scalar product of force and displacement. Positive work: force has a component along the displacement (theta < 90 degrees), e.g. pushing a cart forward increases its kinetic energy. Negative work: force opposes displacement (theta > 90 degrees), e.g. friction or a braking force decreases kinetic energy. Zero work: force is perpendicular to displacement (theta = 90 degrees), e.g. the centripetal force on a satellite, or when there is no displacement at all, e.g. pushing against a wall.

Q7. State and explain the work done by a variable force with its graphical representation.

The work done by a variable force is W = integral of F dx over the displacement, i.e. the area under the force-displacement graph between the initial and final positions. Graphically, the F-x curve is

divided into narrow strips; the area of each strip F dx is summed, which in the limit gives the area under the curve. For a spring, F = k x gives a triangular area (1/2) k x^2, the work done in stretching it.

Q8. State and explain energy with its two forms such as kinetic energy and potential energy. Also describe the work-energy theorem.

Energy is the ability of a body to do work. Its two mechanical forms are: kinetic energy, K.E. = (1/2) m v^2, the energy of a body due to its motion; and potential energy, the energy due to position or configuration, e.g. gravitational P.E. = m g h and elastic P.E. = (1/2) k x^2. The work-energy theorem states that the net work done on a body equals the change in its kinetic energy: W_net = (1/2) m v_f^2 - (1/2) m v_i^2. It follows directly from Newton's second law and the equations of motion.

Exercise - Numerical Problems

Q1. A train moves with a uniform velocity of 24 m s^-1. The driver applies the brake and the train comes to rest with a uniform retardation in 12 s. Find distance covered by train after the brakes are applied. (Book answer: 144 m)

Initial velocity v_i = 24 m s^-1, final velocity v_f = 0, time t = 12 s.

Acceleration a = (v_f - v_i) / t = (0 - 24) / 12 = -2 m s^-2 (retardation 2 m s^-2).

Distance S = v_i t + (1/2) a t^2 = 24 x 12 + (1/2)(-2)(12)^2.

S = 288 - 144 = 144 m.

The distance covered after the brakes are applied is 144 m. This matches the book answer.

Q2. A coin is dropped from a tower. If the coin reaches the ground after 5 s then determine (a) height of the tower and (b) find the speed with which coin hit the ground. (Book answer: (a) 122.5 m (b) 49 m s^-1)

Dropped from rest: v_i = 0, g = 9.8 m s^-2, t = 5 s.

(a) Height h = v_i t + (1/2) g t^2 = 0 + (1/2)(9.8)(5)^2 = 4.9 x 25 = 122.5 m.

(b) Final speed v_f = v_i + g t = 0 + 9.8 x 5 = 49 m s^-1.

(a) Height of the tower 122.5 m; (b) speed on hitting the ground 49 m s^-1. This matches the book answer.

Q3. A 4000 kg truck moving at a speed of 20 m s^-1 along a straight road, strikes a 800 kg stationary car and couples to it. What will be their combined speed after impact? (Book answer: 16.7 m s^-1)

By conservation of momentum: m_truck u_truck + m_car u_car = (m_truck + m_car) V.

4000 x 20 + 800 x 0 = (4000 + 800) V, so 80,000 = 4800 V.

V = 80,000 / 4800 = 16.67 m s^-1, about 16.7 m s^-1.

Their combined speed after impact is 16.7 m s^-1. This matches the book answer.

Q4. Two spheres of masses 'm' and '2m' both are moving to the right with velocities 4 m s^-1 and 2 m s^-1 respectively. If both collide then what will be their final velocities. (Book answer: 1.33 m s^-1 to the right, 4 m s^-1 to the right)
For an elastic collision, v1' = ((m1 - m2) v1 + 2 m2 v2) / (m1 + m2) with m1 = m, m2 = 2m, v1 = 4, v2 = 2.

v1' = ((m - 2m)(4) + 2(2m)(2)) / (3m) = (-4m + 8m) / 3m = 4/3 = 1.33 m s^-1 to the right.

v2' = ((m2 - m1) v2 + 2 m1 v1) / (m1 + m2) = ((2m - m)(2) + 2m(4)) / 3m = (2m + 8m) / 3m = 10/3 = 3.33 m s^-1 to the right.

Check momentum: initial = m(4) + 2m(2) = 8m; final = m(1.33) + 2m(3.33) = 8m. Conserved.

Correct final velocities: 1.33 m s^-1 to the right for mass m, and 3.33 m s^-1 to the right for mass 2m. NOTE: the book prints 4 m s^-1 for the second body, which violates conservation of momentum (1.33m + 8m = 9.33m, not the initial 8m); the values above conserve both momentum (1.33m + 6.67m = 8m) and kinetic energy, as shown in the steps.

Q5. A cricket ball is hit upward at an angle 45 degrees with velocity of 20 m s^-1, find (a) the maximum height (b) time of flight (c) how far away it hits the ground. (Book answer: (a) 10.2 m (b) 2.9 s (c) 41 m)

v_i = 20 m s^-1, theta = 45 degrees, sin theta = cos theta = 0.707, g = 9.8 m s^-2.

(a) H = v_i^2 sin^2 theta / (2g) = 400 x 0.5 / 19.6 = 200 / 19.6 = 10.2 m.

(b) T = 2 v_i sin theta / g = 2 x 20 x 0.707 / 9.8 = 28.28 / 9.8 = 2.89 s, about 2.9 s.

(c) R = v_i^2 sin 2 theta / g = 400 x sin 90 degrees / 9.8 = 400 / 9.8 = 40.8 m, about 41 m.

(a) Maximum height 10.2 m; (b) time of flight 2.9 s; (c) range 41 m. This matches the book answer.

Q6. In certain projectile motion, the horizontal range is thrice the magnitude of the maximum height. Calculate the angle of projection. (Book answer: 53 degrees)

R = v_i^2 sin 2 theta / g = v_i^2 (2 sin theta cos theta) / g.

H = v_i^2 sin^2 theta / (2g). Given R = 3H.

2 sin theta cos theta = 3 sin^2 theta / 2, so dividing by sin theta: 2

cos theta = (3/2) sin theta.

tan theta = 4/3, so theta = tan^-1(1.333) = 53.1 degrees, about 53 degrees.

The angle of projection is 53 degrees. This matches the book answer.

Q7. Prove that the horizontal ranges in projectile motion are same at (45 degrees + theta) and (45 degrees - theta).
Range R = v_i^2 sin 2(phi) / g, where phi is the projection angle.

For phi = 45 degrees + theta: 2 phi = 90 degrees + 2 theta, and sin(90 degrees + 2 theta) = cos 2 theta.

For phi = 45 degrees - theta: 2 phi = 90 degrees - 2 theta, and sin(90 degrees - 2 theta) = cos 2 theta.

Both give R = v_i^2 cos 2 theta / g, hence the ranges are equal.

The ranges are equal. Proved.

Q8. A 250 kg cart is pushed up on an inclined surface. How much work does the pushing force when the cart moves up and at 3 m above the ground, friction is neglected? (Book answer: 7350 J)
With friction neglected, the work of the pushing force equals the gain in gravitational potential energy.

Gain in P.E. = m g h = 250 x 9.8 x 3.

250 x 9.8 = 2450; 2450 x 3 = 7350 J.

The pushing force does 7350 J of work. This matches the book answer.

Q9. A force of 6000 N is applied horizontally on a van of mass 2500 kg. The van starts its motion from rest and if it has traveled a distance of 110 m. What will be its speed and K.E.? (Book answer: 23 m s^-1, 660 kJ)
Acceleration a = F / m = 6000 / 2500 = 2.4 m s^-2.

From rest, v^2 = v_i^2 + 2 a S = 0 + 2 x 2.4 x 110 = 528.

v = sqrt(528) = 22.98 m s^-1, about 23 m s^-1.

K.E. = (1/2) m v^2 = (1/2) x 2500 x 528 = 660,000 J = 660 kJ.

Speed 23 m s^-1 and K.E. 660 kJ. This matches the book answer.

Q10. How much work is required to accelerate a body of mass 200 kg from 5 m s^-1 to 15 m s^-1? If it covers a distance of 150 m what is the net force acting on it. (Book answer: 2 x 10^4 J, 133 N)
By the work-energy theorem, W = change in K.E. = (1/2) m (v_f^2 - v_i^2).

W = (1/2) x 200 x (225 - 25) = 100 x 200 = 20,000 J = 2 x 10^4 J.

Net force F = W / S = 20,000 / 150 = 133.3 N, about 133 N.

Work required 2 x 10^4 J and net force 133 N. This matches the book answer.

Unit 4: Rotational Motion

Exercise - Multiple Choice Questions (MCQs)

Q1. 85.95 degree in terms of radian is:

- 1/2 radian
- 1 radian
- ✓ 1 1/2 radian
- 2 radian

Q2. The circumference of a circle of radius 50 cm is:

- 3.12 m
- ✓ 3.14 m
- 3.16 m
- 3.18 m

Q3. A body moves with constant angular velocity in a circle. The magnitude of its angular acceleration is:

- r omega^2
- Constant
- ✓ zero
- r omega

Q4. A solid cylinder of mass 20 kg rotates about its axis with angular velocity of 10 m s^-1 and radius 0.2 m. The moment of inertia of the cylinder is:

- 0.2 kg m^2

✓ 0.4 kg m^2

0.6 kg m^2

0.8 kg m^2

Q5. The condition of weightlessness is (N = Normal Reaction and W is weight):

N > W

N < W

N = W

✓ N = 0

Q6. The frequency of artificial gravity is given by:

2 pi sqrt(g / R)

2 pi sqrt(R / g)

(1 / 2 pi) sqrt(R / g)

✓ (1 / 2 pi) sqrt(g / R)

Q7. A particle of mass 100 g starts its motion from rest along a circular path of radius 10 cm. If its velocity becomes 10 m s^-1 then its centripetal force is:

0.1 N

1 N

10 N

✓ 100 N

Q8. Which one of the following forces produces a centripetal acceleration in a revolving satellite?

Friction

Normal

Tension

✓ Gravitation

Exercise - Short Questions

Q1. How many units are being used for measurement of angular displacement? Explain.

Three units are used: (i) the degree (360 degrees = one revolution), (ii) the radian (the angle subtended when arc length equals radius; 2 pi radians = one revolution), and (iii) the revolution itself (one complete turn). The radian is the SI unit; degrees are convenient for everyday use.

Q2. What is the relationship between arc length and angular displacement?

Arc length S and angular displacement theta are related by S = r theta, where r is the radius of the circle and theta is measured in radians. Thus a larger radius or a larger angle gives a proportionally longer arc.

Q3. How can a body move in a circle? What are the direction of its linear and angular velocities?

A body moves in a circle when a centripetal force acts on it continuously towards the centre, constantly changing the direction of its velocity without changing its speed. Its linear (tangential) velocity is directed along the tangent to the circle at each point, while its angular velocity is directed along the axis of rotation, by the right-hand rule.

Q4. Why torque and work done are zero by centripetal force?

The centripetal force always acts towards the centre, perpendicular to the tangential displacement of the body, so the work done W = F S cos 90 degrees = 0. Its torque about the centre is also zero because torque = r x F, and the position vector r (outward) and the force F (inward) are anti-parallel, giving a zero cross product. Hence the centripetal force changes direction only, never speed or angular momentum.

Q5. There are two spheres of copper and lead having same mass. It is found that the lead sphere can be rotated more easily. Explain why?

Lead is much denser than copper, so a lead sphere of the same mass has a smaller radius. Moment of inertia I = (2/5) M R^2 for a solid sphere depends on R^2, so the lead sphere has a smaller moment of inertia and offers less resistance to rotation; hence it can be rotated more easily.

Q6. The cylinders A and B are of the same mass but the radius of A is greater than that of B. Which one will require more force to come into rotation? Why?

Cylinder A will require more force (torque) to come into rotation. Both have the same mass, but moment of inertia increases with radius (for a cylinder I = (1/2) M R^2), so A has the larger moment of inertia and offers greater resistance to angular acceleration; by tau = I alpha, a larger torque is needed for the same angular acceleration.

Q7. How the rotation of Earth will be affected if its density becomes uniform?

The Earth is denser at its core than at its surface, so its moment of inertia is about 0.33 M R^2, less than the 0.4 M R^2 of a uniform sphere. If its density became uniform (same mass and radius), its moment of inertia would increase, and by conservation of angular momentum (L = I omega constant, no external torque) its angular velocity would decrease: the Earth would spin more slowly and the day would become longer.

Exercise - Comprehensive Questions

Q1. Derive the relationship between linear and angular variables.

The linear and angular variables are related by: S = r theta (arc length = radius x angular displacement), v = r omega (tangential speed = radius x angular velocity), and a_t = r alpha (tangential acceleration = radius x angular acceleration). In addition, the centripetal acceleration is a_c = v^2 / r = r omega^2, directed towards the centre. These follow by differentiating S = r theta with respect to time, since r is constant for circular motion.

Q2. State and explain centripetal acceleration and centripetal force. Derive the formula for centripetal force.

Centripetal acceleration is the acceleration of a body in circular motion, directed towards the centre, of magnitude a_c = v^2 / r = r omega^2. It arises because the direction of velocity changes continuously. The force producing it is the centripetal force F_c = m a_c = m v^2 / r = m r omega^2. Derivation: consider a body

moving with constant speed v on a circle of radius r; in a short time the velocity vector turns through the same angle as the radius, and from the similar triangles formed by the displacement and velocity changes, delta v / v = delta S / r; dividing by delta t and taking the limit gives a = v^2 / r towards the centre, hence F_c = m v^2 / r.

Q3. What is moment of inertia? Show that moment of inertia depends upon its mass and radius of the circle.

Moment of inertia is the rotational analogue of mass: the resistance of a body to change in its rotational motion, I = sum of m r^2 over all particles, where r is the distance from the axis. For a single particle I = m r^2, showing direct dependence on mass and on the square of the distance (radius). For extended bodies the same dependence appears: a hoop I = M R^2, a disc I = (1/2) M R^2, a solid sphere I = (2/5) M R^2, a rod about its centre I = (1/12) M L^2. In every case I grows with mass and with the square of the size, which is why mass far from the axis contributes most.

Q4. State and explain angular momentum and law of conservation of angular momentum.

Angular momentum is L = r x p = I omega, the vector product of position vector and linear momentum; for rotation about a fixed axis its magnitude is L = I omega. The law of conservation of angular momentum states: if no external torque acts on a system, its total angular momentum remains constant. Examples: an ice skater spins faster when she pulls her arms in (I decreases, so omega increases to keep I omega constant), and a diver tucks to increase spin rate.

Q5. What do you know about the weightlessness in satellite? Describe the conditions of weightlessness.

Weightlessness in a satellite means the apparent weight (normal reaction) is zero, although gravity still acts. An astronaut in orbit is in continuous free fall around the Earth: both satellite and astronaut fall towards the Earth with the same centripetal acceleration g, so no normal force acts between them and they feel weightless. Conditions for weightlessness: (i) the body must be in free fall (only gravity acting), (ii) the normal reaction N = 0, and (iii) the body and its support must share the same acceleration, as in an orbiting satellite or a freely falling lift.

Q6. How can you produce the artificial gravity? Calculate the frequency for artificial gravity.

Artificial gravity is produced by rotating a spacecraft about its own axis: the wall of the rotating rim pushes the astronaut towards the centre, providing the centripetal force which the astronaut feels as weight. For artificial gravity equal to g, the centripetal acceleration must satisfy v^2 / R = g, so v = sqrt(g R), and the frequency of rotation is f = v / (2 pi R) = (1 / 2 pi) sqrt(g / R). For example, a spacecraft of radius 60 m needs v = sqrt(9.8 x 60) = 24.2 m s^-1 and a time period T = 2 pi R / v = 15.6 s (see Numerical Q4).

Exercise - Numerical Problems

Q1. The Earth completes one rotation about its axis in 24 hours. Calculate (a) the angular speed of the Earth (b) tangential speed of body at the equator. (Radius of the Earth is 6.4 x 10^6 m) (Book

answer: 7.3 x 10^-5 rad/s, 467 m s^-1)

Time period T = 24 hours = 24 x 3600 = 86,400 s.

(a) Angular speed omega = 2 pi / T = 6.2832 / 86,400 = 7.27 x 10^-5 rad/s, about 7.3 x 10^-5 rad/s.

(b) Tangential speed v = omega R = 7.27 x 10^-5 x 6.4 x 10^6 = 465 m s^-1; using the rounded omega 7.3 x 10^-5 gives 467 m s^-1.

(a) Angular speed 7.3 x 10^-5 rad/s; (b) tangential speed 467 m s^-1. This matches the book answer.

Q2. A geostationary satellite is revolving around the Earth in the orbit of radius 4.24 x 10^7 m in time period of 24 hours. Calculate (a) tangential speed (b) centripetal acceleration. (Book answer: 3.08 x 10^3 m s^-1, 0.224 m s^-2)

Time period T = 24 x 3600 = 86,400 s; radius R = 4.24 x 10^7 m.

(a) v = 2 pi R / T = 2 x 3.1416 x 4.24 x 10^7 / 86,400.

Numerator = 2.664 x 10^8; dividing by 86,400 gives v = 3083 m s^-1, about 3.08 x 10^3 m s^-1.

(b) Centripetal acceleration a = v^2 / R = (3083)^2 / (4.24 x 10^7) = 9.50 x 10^6 / 4.24 x 10^7 = 0.224 m s^-2.

(a) Tangential speed 3.08 x 10^3 m s^-1; (b) centripetal

acceleration 0.224 m s^-2. This matches the book answer.

Q3. A 3 kg pulley of radius 30 cm is rotating at the rate of 400 rev/min. Calculate its moment of inertia and its rotational K.E. (Book answer: 0.27 kg m^2, 548 kJ)

Radius R = 30 cm = 0.30 m; mass M = 3 kg.

For a pulley with mass concentrated at the rim, I = M R^2 = 3 x (0.30)^2 = 3 x 0.09 = 0.27 kg m^2.

Angular velocity omega = 400 rev/min = 400 x 2 pi / 60 = 41.89 rad/s.

Rotational K.E. = (1/2) I omega^2 = (1/2) x 0.27 x (41.89)^2 = 0.135 x 1754.8 = 237 J.

Moment of inertia 0.27 kg m^2 (rim model, I = M R^2) and rotational K.E. 237 J. NOTE: the book prints rotational K.E. 548 kJ, which cannot be reproduced from the printed data: with omega = 41.9 rad/s and I = 0.27 kg m^2, (1/2) I omega^2 = 237 J; no standard conversion of 400 rev/min gives 548 kJ. The step-by-step value 237 J is used here.

Q4. At what speed of the outer rim of space craft is rotated in order to produce an artificial gravity equal to 9.8 m s^-2? The radius of space craft is 60 m, also calculate its time period of rotation. (Book answer: 42.2 m s^-1, 15.6 s)

Artificial gravity requires centripetal acceleration v^2 / R = g = 9.8

m s^-2.

v^2 = g R = 9.8 x 60 = 588, so v = sqrt(588) = 24.25 m s^-1, about 24.2 m s^-1.

Time period T = 2 pi R / v = 2 x 3.1416 x 60 / 24.25 = 376.99 / 24.25 = 15.5 s, about 15.6 s.

Correct rim speed 24.2 m s^-1 with time period 15.6 s. NOTE: the book prints 42.2 m s^-1, which is inconsistent with its own period: v = 42.2 m s^-1 would give T = 2 pi x 60 / 42.2 = 8.9 s, not 15.6 s; the pair (24.2 m s^-1, 15.6 s) is mutually consistent, as shown below.

Unit 5: Physics of Solids

Exercise - Multiple Choice Questions (MCQs)

Q1. Copper is an example of:

- ✓ Crystal
- Amorphous
- Polymer
- Ceramic

Q2. By increasing the temperature, the distance between the molecules of a crystal is:

- Decreased
- ✓ Increased
- Same
- Zero

Q3. Which one of the following is called glassy solid?

- Crystal
- ✓ Amorphous
- Polymer
- Diamond

Q4. Hook's law provides the relationship between:

- Force and weight
- Force and pressure
- ✓ Stress and Strain

Displacement and dimension

Q5. The property of a body to attain its original position after the removing of force is called:

Plasticity

Rigidity

✓ Elasticity

Resistivity

Q6. Which one of the following quantity has no dimension and unit?

Stress

✓ Strain

Tangential Stress

Young Modulus

Q7. A body which breaks down just crossing the elastic limit is known as:

Elastic

Plastic

Ductile

✓ Brittle

Q8. When a change is produced in a body in its shape only then its corresponding strain is called:

Tensile

Compression

Volumetric

✓ Shearing

Exercise - Short Questions

Q1. Why does an amorphous solid has the same property as that of a frozen liquid?

An amorphous solid has no regular arrangement of its atoms or molecules; its structure is disordered like that of a liquid, only frozen in place. It therefore flows extremely slowly, has no sharp melting point, and breaks with an irregular fracture, just like a frozen liquid. Glass is the classic example.

Q2. Why is amorphous also called glassy solids?

Amorphous solids are called glassy solids because glass is the best-known example: like glass, they have a disordered, non-crystalline structure, no definite melting point (they soften gradually on heating), and they break with a conchoidal (shell-like) fracture.

Q3. Is stress different from pressure? If no then explain it briefly.

Yes and no: stress and pressure have the same unit (N m^-2, pascal) and the same dimensions, but they are different quantities. Pressure is the normal force per unit area exerted by a fluid in all directions, while stress is the internal restoring force per unit area developed inside a solid, and it can be tensile, compressive or shearing, not only normal.

Q4. What is meant by the co-efficient of elasticity?

The coefficient of elasticity (elastic modulus) is the ratio of stress to strain within the elastic limit: modulus = stress / strain. It measures

the stiffness of a material: a larger modulus means a larger stress is needed to produce a given strain. Its three kinds are Young's modulus (longitudinal), bulk modulus (volumetric) and shear modulus (tangential).

Q5. What do you know about the ultimate strength of a solid?
The ultimate strength of a solid is the maximum stress it can withstand before breaking (fracture). On a stress-strain graph it is the highest point of the curve; beyond it the material necks down and fractures. It is the limit of load-bearing capacity of the material.

Q6. Which is more elastic rubber or iron? Explain briefly.
Iron is more elastic than rubber. Elasticity is measured by the elastic modulus (stress/strain), not by how far a material stretches: iron needs a very large stress to produce a small strain (Young's modulus about 2 x 10^11 Pa), while rubber stretches easily under a small stress (modulus about 10^7 Pa). Rubber is more stretchable, but iron is more elastic in the scientific sense.

Q7. Differentiate between elasticity and plasticity of solids.
Elasticity is the property of a body to regain its original shape and size completely when the deforming force is removed, provided the elastic limit is not crossed. Plasticity is the property by which a body retains a permanent deformation after the force is removed: the change in shape or size remains even when the force is gone. A spring shows elasticity; a bent copper wire shows plasticity.

Q8. Distinguish between ultimate tensile strength (UTS) and yield

tensile strength (YTS).

Ultimate tensile strength (UTS) is the maximum tensile stress a material can bear before fracture; it is the peak of the stress-strain curve. Yield tensile strength (YTS) is the stress at which the material stops behaving elastically and begins to deform permanently (the yield point); beyond it, removing the load leaves a permanent strain. UTS is always greater than YTS.

Exercise - Comprehensive Questions

Q1. Discuss the three classes of solids: crystal, amorphous and polymeric with examples.

The three classes of solids are: (i) Crystalline solids: atoms, molecules or ions arranged in a regular repeating pattern, e.g. copper, iron, diamond, common salt; they have sharp melting points and definite cleavage planes. (ii) Amorphous solids: atoms arranged randomly with no long-range order, e.g. glass, plastics, rubber; they soften over a range of temperatures and have no sharp melting point. (iii) Polymeric solids: long chains of repeated molecular units, neither perfectly regular nor fully random, e.g. polythene, nylon, PVC; they are flexible, light and can be moulded.

Q2. What is meant by the mechanical properties of solids? Explain it with examples.

Mechanical properties describe how a solid responds to applied forces: elasticity (regaining shape after deformation), plasticity (permanent deformation), strength (maximum stress before fracture), stiffness (modulus), hardness (resistance to scratching or indentation), ductility (ability to be drawn into wires, e.g. copper),

brittleness (breaking without appreciable deformation, e.g. glass), and toughness (energy absorbed before fracture). For example, steel is strong, stiff and ductile, while glass is hard but brittle.

Q3. State and explain stress, strain and their kinds with suitable examples.

Stress is the internal restoring force per unit area, stress = F / A (unit pascal). Its kinds: tensile stress (pulling, increasing length), compressive stress (pushing, decreasing length) and tangential or shear stress (forces parallel to the surface, changing shape). Strain is the fractional deformation, strain = change in dimension / original dimension (dimensionless). Its kinds: longitudinal (tensile or compressive) strain = change in length / original length, volumetric strain = change in volume / original volume, and shear strain = angular change (in radians) produced by tangential stress.

Q4. Define Hooke's law in terms of stress and strain. Also discuss the three kinds of elastic moduli.

Hooke's law in terms of stress and strain states: within the elastic limit, stress is directly proportional to strain, stress = (constant) x strain. The constant is the elastic modulus. Its three kinds are: (i) Young's modulus Y = longitudinal stress / longitudinal strain, (ii) bulk modulus K = volumetric stress / volumetric strain (with a minus sign, since volume decreases as pressure increases), and (iii) shear modulus G = tangential stress / shear strain. Each measures stiffness against a different kind of deformation.

Q5. Explain graphical representation of Hooke's law and define the

terms: Elastic limit, plastic limit, yield strength and ultimate strength.

On a stress-strain graph, stress rises linearly with strain up to the proportional limit (Hooke's law region, a straight line), then curves to the elastic limit: the maximum stress from which the body fully recovers. Beyond it lies the plastic region, where permanent deformation occurs. The yield strength (yield point) is the stress at which plastic deformation begins clearly; the ultimate strength is the peak stress before the curve falls to the fracture point where the body breaks. The area under the curve up to fracture measures toughness.

Exercise - Numerical Problems

Q1. An iron rod 4 m long and 0.5 cm^2 in cross section area stretches by 0.5 mm when a mass of 115 kg is hung from its lower end. Compute stress, strain and Young's modulus. (Book answer: 2.254 x 10^11 Pa, 1.25 x 10^-4, 1.8 x 10^11 Pa)

Force F = m g = 115 x 9.8 = 1127 N.

Area A = 0.5 cm^2 = 0.5 x 10^-4 m^2 = 5 x 10^-5 m^2.

Stress = F / A = 1127 / (5 x 10^-5) = 2.254 x 10^7 Pa.

Strain = change in length / original length = 0.5 x 10^-3 / 4 = 1.25 x 10^-4.

Young's modulus Y = stress / strain = 2.254 x 10^7 / 1.25 x 10^-4 = 1.8 x 10^11 Pa.

Stress 2.254 x 10^7 Pa, strain 1.25 x 10^-4, Young's modulus 1.8 x 10^11 Pa. NOTE: the book prints stress as 2.254 x 10^11 Pa, which is 10,000 times too large; the correct stress 2.254 x 10^7 Pa divided by the strain gives exactly the book's Young's modulus 1.8 x 10^11 Pa, confirming the misprint.

Q2. A solid cylindrical steel column is 6 m long and 10 cm is diameter. What will be its increase in two length when carrying a load of 2 x 10^5 kg (Young Modulus for steel is 1.9 x 10^11 Pa) (Book answer: 7.88 mm)
Load F = m g = 2 x 10^5 x 9.8 = 1.96 x 10^6 N.

Radius r = 10 cm / 2 = 5 cm = 0.05 m; area A = pi r^2 = 3.1416 x (0.05)^2 = 7.854 x 10^-3 m^2.

Increase in length delta L = F L / (A Y) = 1.96 x 10^6 x 6 / (7.854 x 10^-3 x 1.9 x 10^11).

Numerator = 1.176 x 10^7; denominator = 1.492 x 10^9.

delta L = 7.88 x 10^-3 m = 7.88 mm.

Increase in length 7.88 mm. This matches the book answer.

Q3. Two parallel and opposite forces, each 4000 N are applied tangentially to the upper and lower faces of cubical metal block 25 cm on a side. Find the angle of shear. (Shearing modulus is 80 GPa)

(Book answer: 8 x 10^-5 rad.)

Face area A = (0.25 m)^2 = 0.0625 m^2.

Shear stress = F / A = 4000 / 0.0625 = 64,000 Pa = 6.4 x 10^4 Pa.

Shear modulus G = 80 GPa = 80 x 10^9 Pa.

Angle of shear theta = shear stress / G = 6.4 x 10^4 / (80 x 10^9) = 8 x 10^-7 rad.

Angle of shear 8 x 10^-7 rad. NOTE: the book prints 8 x 10^-5 rad, which is 100 times too large; the step-by-step division below gives 8 x 10^-7 rad.

Q4. Calculate change in volume of a solid cube 50 mm on each edge, when subjected to a stress of 25 MPa. (The bulk modulus for copper is 125 GPa) (Book answer: 25 mm^3)

Original volume V = (50 mm)^3 = 125,000 mm^3.

Volumetric strain = stress / bulk modulus = 25 x 10^6 / (125 x 10^9) = 2 x 10^-4.

Change in volume delta V = V x volumetric strain = 125,000 x 2 x 10^-4 = 25 mm^3.

Change in volume 25 mm^3. This matches the book answer.

Unit 6: Fluid Dynamics

Exercise - Multiple Choice Questions (MCQs)

Q1. Principle of floatation in terms of upthrust force F_u and weight of the body 'W' is:

- F_u > W
- F_u = W
- F_u < W
- ✓ (a) and (b)

Q2. Viscosity of a fluid depends upon:

- Mass
- Density
- Volume
- ✓ Temperature

Q3. Drag force exerted by the fluid on a body does not depend upon:

- Viscosity of fluid
- Terminal velocity
- Shape of a body
- ✓ Volume of fluid

Q4. In incompressible fluid, which parameter remains constant?

- Pressure
- Volume

Temperature

✓ Density

Q5. Equation of continuity has been derived on the basis of law of conservation of:

Energy

Momentum

✓ Mass

Force

Q6. According to equation of continuity A1 V1 = A2 V2 = constant. The constant is equal to:

✓ Flow rate

Volume of fluid

Mass of fluid

Density of fluid

Q7. Bernoulli's equation has been derived on the basis of conservation of:

Mass

✓ Energy

Momentum

Force

Q8. A two-metre high tank is full of water. If a hole appears at its middle, then the speed of efflux is:

✓ 4.42 m s^-1

5.42 m s^-1

6.42 m s^-1

7.42 m s^-1

Q9. Venturi-meter is an instrument which is being used for the measurement of:

Density of fluid

✓ Velocity of fluid

Pressure of fluid

Viscosity of fluid

Q10. When velocity of fluid is increased then its pressure is:

Increased

✓ Decreased

Same

Not velocity dependent

Exercise - Short Questions

Q1. What do you know about the upthrust force of a liquid?

Upthrust (buoyant force) is the upward force exerted by a liquid on a body immersed in it. By Archimedes' principle it equals the weight of the liquid displaced by the body: F_u = (volume immersed) x (density of liquid) x g. It acts vertically upward through the centre of buoyancy.

Q2. How does a heavy ship float in water?

A heavy ship floats because its hull encloses a large volume of air, so its average density (total mass divided by total volume including the hollow hull) is less than the density of water. The upthrust on

the large displaced volume of water then equals the ship's weight, satisfying the law of floatation, even though the steel itself is denser than water.

Q3. Does a non viscous fluid exist? If yes, how a fluid be made non-viscous.

A perfectly non-viscous (ideal) fluid does not exist in nature; all real fluids have some viscosity. A fluid can be made effectively non-viscous only in special states: liquid helium below 2.17 K becomes a superfluid with zero viscosity, and gases at very low density behave almost ideally. In textbook problems, water is often treated as nearly non-viscous as an approximation.

Q4. Why do raindrops not hurt us?

Raindrops do not hurt us because they quickly reach a small terminal velocity. As a drop falls, air drag grows with speed until it balances the drop's weight; the drop then falls at constant speed (about 9 m s^-1 for a large drop) instead of accelerating all the way down. Their small mass also means their momentum on impact is tiny.

Q5. How does a steady flow differs from turbulent flow?

In steady (streamline or laminar) flow, every particle passing a given point follows the same path with the same velocity, and the flow pattern does not change with time. In turbulent flow the velocity at a point keeps changing in magnitude and direction, eddies and irregular whirls form, and the motion is disorderly. Steady flow occurs at low speeds; above a critical speed flow

becomes turbulent.

Q6. Why is liquid incompressible while gas is compressible?

In a liquid the molecules are already close together with strong intermolecular forces, so squeezing them closer is very difficult: the volume hardly changes under pressure. In a gas the molecules are far apart with large empty spaces between them, so pressure can push them much closer together and the volume changes easily. Hence liquids are practically incompressible while gases are compressible.

Q7. How is variation in pressure affected the speed of fluid?

By Bernoulli's principle, for horizontal flow an increase in speed is accompanied by a decrease in pressure, and a decrease in speed by an increase in pressure: P + (1/2) rho v^2 = constant. So where a fluid speeds up (e.g. over an aeroplane wing or through a constriction), its pressure falls; where it slows down, its pressure rises.

Exercise - Comprehensive Questions

Q1. State and explain Archimedes's principle with examples.

Archimedes' principle states: when a body is wholly or partially immersed in a fluid, it experiences an upward buoyant force (upthrust) equal to the weight of the fluid displaced by it. Examples: a ship floats because the weight of water it displaces equals its weight; a hydrometer floats higher in denser liquids because less volume must be displaced to balance its weight; a balloon rises because the upthrust of displaced air exceeds its

weight.

Q2. What do you know about the law of floatation? Explain the conditions of law of floatation.

The law of floatation states: a body floats in a fluid when the upthrust equals its weight, i.e. when the weight of the displaced fluid equals the weight of the body. Conditions: (i) if the average density of the body is less than the fluid density, it floats (partly immersed); (ii) if it equals the fluid density, it stays suspended anywhere in the fluid; (iii) if it is greater, the body sinks, since the maximum upthrust (fully immersed) is still less than its weight. A floating body displaces its own weight of fluid.

Q3. Define fluid and describe the viscosity of a fluid. Also express the relation for viscosity of fluid.

A fluid is a substance that flows and takes the shape of its container; liquids and gases are fluids. Viscosity is the internal friction of a fluid: the resistance between its layers to relative motion, arising from intermolecular forces and momentum transfer between layers. It is measured by the coefficient of viscosity eta, defined by Newton's law of viscosity: F = eta A (dv/dy), i.e. eta = F / (A x velocity gradient), with SI unit N s m^-2 (pascal-second). Liquids like honey have high viscosity; water has low viscosity, and viscosity of liquids decreases with temperature.

Q4. State and explain the terminal velocity with the help of Stokes' law.

Terminal velocity is the constant maximum velocity attained by a

body falling through a fluid when the upward drag plus upthrust exactly balance its weight, so the net force and acceleration become zero. By Stokes' law, the drag on a small sphere moving slowly through a viscous fluid is F = 6 pi eta r v, proportional to viscosity, radius and speed. Equating forces (weight = upthrust + drag) gives the terminal velocity v_t = (2 r^2 g (rho_body - rho_fluid)) / (9 eta); it grows with the square of the radius and with the density difference, and falls as viscosity rises. Raindrops and parachutists are everyday examples.

Q5. State equation of continuity and derive equation of continuity on the basis of conservation of mass.

The equation of continuity states: for steady, incompressible flow, A v = constant along a pipe, where A is the cross-sectional area and v the flow speed. Derivation from conservation of mass: the mass entering a section per second, rho A1 v1, must equal the mass leaving per second, rho A2 v2, since mass is neither created nor destroyed; for an incompressible fluid rho cancels, giving A1 v1 = A2 v2. Hence the fluid speeds up where the pipe narrows, as when the end of a garden hose is squeezed.

Q6. State and explain Bernoulli's equation and derive it on the basis of law of conservation of energy.

Bernoulli's equation states: for steady, incompressible, non-viscous flow, P + (1/2) rho v^2 + rho g h = constant along a streamline. Derivation from conservation of energy: the work done by pressure forces on a fluid element equals the change in its kinetic and potential energy; dividing by the volume gives the pressure term,

the kinetic-energy term (1/2) rho v^2 and the potential term rho g h, whose sum stays constant. Applications include aeroplane lift, the Venturi meter, atomizers and the swinging of a spinning cricket ball.

Q7. What do you know about the superfluid and super fluidity?

A superfluid is a liquid phase, notably liquid helium-4 below 2.17 K (the lambda point), in which viscosity drops to zero. Superfluidity is this state of zero viscosity: a superfluid flows through the narrowest capillaries without friction, creeps up and over the walls of its container (film flow), and once set rotating keeps its vortices spinning indefinitely. It is a quantum phenomenon explained by a fraction of the atoms occupying the lowest energy state.

Q8. Discuss the construction and working principle of filter pump and atomizer.

A filter pump (Venturi pump) works on Bernoulli's principle: water forced through a constriction speeds up and its pressure falls below atmospheric, so air is sucked in through a side tube and carried away, producing a partial vacuum for filtration. An atomizer (spray bottle) works the same way: air blown fast across the top of a vertical tube lowers the pressure there, atmospheric pressure pushes the liquid up the tube, and the fast air stream breaks it into a fine spray.

Exercise - Numerical Problems

Q1. Calculate the average speed of water flowing through a pipe of diameter 10 cm which delivers 5 m^3 of water per hour. (Book

answer: 0.18 m s^-1)

Volume flow rate Q = 5 m^3 per hour = 5 / 3600 = 1.389 x 10^-3 m^3 s^-1.

Pipe radius r = 10 cm / 2 = 5 cm = 0.05 m; area A = pi r^2 = 3.1416 x (0.05)^2 = 7.854 x 10^-3 m^2.

Average speed v = Q / A = 1.389 x 10^-3 / 7.854 x 10^-3 = 0.177 m s^-1, about 0.18 m s^-1.

Average speed 0.18 m s^-1. This matches the book answer.

Q2. The speed of water 0.5 m s^-1 flowing in a pipe of diameter 5 cm. What will be its speed in a pipe of 2.5 cm diameter that is connected with it. (Book answer: 2 m s^-1)

By the equation of continuity, A1 v1 = A2 v2.

Area is proportional to diameter squared: A2 / A1 = (2.5 / 5)^2 = (1/2)^2 = 1/4.

Hence v2 = v1 x (A1 / A2) = 0.5 x 4 = 2 m s^-1.

Speed in the narrower pipe 2 m s^-1. This matches the book answer.

Q3. What will be the gauge pressure in a large fire hose if the nozzle is to shoot water straight upward to a height of 25 m? (Book answer: 2.45 x 10^5 Pa)

The pressure must support a 25 m column of water: P = rho g h.

P = 1000 x 9.8 x 25 = 245,000 Pa.

P = 2.45 x 10^5 Pa.

Gauge pressure 2.45 x 10^5 Pa. This matches the book answer.

Q4. What is the height of water inside the tank above the orifice if the velocity of efflux of water through orifice is 9.9 m s^-1. (Book answer: 5 m)

Torricelli's theorem: velocity of efflux v = sqrt(2 g h).

So h = v^2 / (2g) = (9.9)^2 / (2 x 9.8) = 98.01 / 19.6 = 5.0 m.

Height of water above the orifice 5 m. This matches the book answer.

Q5. A liquid of density 8 x 10^3 kg m^-3 is flowing through a horizontal pipe of different cross section. If the pressure difference between two points is 4 x 10^4 N m^-2 then what is the speed of liquid in the tube. (Book answer: 3.16 m s^-1)

For horizontal flow Bernoulli's equation gives delta P = (1/2) rho v^2 for the speed associated with the pressure difference.

v^2 = 2 delta P / rho = 2 x 4 x 10^4 / (8 x 10^3) = 80,000 / 8000 = 10.

v = sqrt(10) = 3.16 m s^-1.

Speed of the liquid 3.16 m s^-1. This matches the book answer.

Q6. An airplane wing is designed such that speed of the air across the top of the wing is 450 m s^-1 and the speed of the air below the wing is 410 m s^-1. What is the pressure difference between the top and the bottom of the wings? (Density of air is 1.29 kg m^-3) (Book answer: 22 kPa)

By Bernoulli's equation for horizontal flow, delta P = (1/2) rho (v_top^2 - v_bottom^2).

v_top^2 = 450^2 = 202,500; v_bottom^2 = 410^2 = 168,100; difference = 34,400.

delta P = (1/2) x 1.29 x 34,400 = 0.645 x 34,400 = 22,188 Pa, about 22 kPa.

Pressure difference 22 kPa. This matches the book answer.

Unit 7: Thermodynamics

Exercise - Multiple Choice Questions (MCQs)

Q1. The equation of state is being applied for a:

- Real gas
- ✓ Ideal gas
- Diatomic gas
- Polyatomic gas

Q2. When two bodies are made at thermal contact having the same temperature then they are at:

- Physical equilibrium
- ✓ Thermal equilibrium
- Mechanical equilibrium
- Chemical equilibrium

Q3. When the system is expanded by adding heat energy then the work done is:

- Positive and on the system
- Negative and on the system
- ✓ Positive and by the system
- Negative and by the system

Q4. Internal energy of a substance is defined in terms of:

- P and V
- P and T

✓ T and V

P, V and T

Q5. The ratio between work done and heat energy is equal to:

Adiabatic constant

✓ Joule's constant

Specific heat constant

Real gas constant

Q6. First law of thermodynamics is based upon law of conservation of:

Mass

✓ Energy

Momentum

Charges

Q7. The efficiency of a heat engine will be 100% when:

Engine takes huge amount of heat from source

Engine exhaust a very small amount of heat from sink

The temperature of cold reservoir is 0 degrees C

✓ The temperature of cold reservoir is 0 K

Q8. All the supplied heat energy is used for useful work in the process of:

Isochoric

Isobaric

✓ Isothermal

Adiabatic

Q9. Second law of thermodynamics provides the proper direction of:

- Temperature
- Force
- Pressure
- ✓ Flow of heat

Q10. Entropy of a reversible process:

- Decreases
- Increases
- Infinite
- ✓ Zero

Exercise - Short Questions

Q1. What is the condition of perfect thermal equilibrium?

Perfect thermal equilibrium means two bodies in thermal contact have exactly the same temperature throughout, so no net heat flows between them. In addition, for complete thermodynamic equilibrium, mechanical equilibrium (equal pressures) and chemical equilibrium (no net chemical change) must also hold.

Q2. Why is the Earth not in thermal equilibrium with the Sun?

The Earth is not in thermal equilibrium with the Sun because their temperatures differ enormously (Sun's surface about 5800 K, Earth's average about 288 K) and heat flows continuously from the Sun to the Earth. The Earth also radiates heat back to space, so the two are never at a common temperature; equilibrium would require

no net heat flow.

Q3. Why does the general gas equation not apply for real gas?

The general gas equation PV = nRT assumes gas molecules are point masses with no volume and no intermolecular forces. Real gas molecules have finite size and attract each other, so at high pressure and low temperature their behaviour deviates from PV = nRT; the van der Waals equation, with corrections for molecular volume and attraction, is needed instead.

Q4. The average kinetic energy of a gas molecules proportional to which state variables?

The average kinetic energy of a gas molecule is proportional to the absolute temperature of the gas: average K.E. = (3/2) k T for a monatomic gas, where k is Boltzmann's constant. It does not depend on pressure or volume separately, only on T.

Q5. What is the difference between the work done on the system and work done by the system?

Work done on the system is positive when an external agent compresses the system (energy enters the system, volume decreases), while work done by the system is positive when the system expands against an external pressure (energy leaves the system, volume increases). In the physics sign convention used in this book, work done by the system during expansion is taken as positive, and work done on the system during compression is negative; the two are equal in magnitude and opposite in sign for the same change.

Q6. Why the adiabatic curve is steeper than the isothermal curve?

The adiabatic curve is steeper than the isothermal curve because in an adiabatic change no heat enters or leaves, so expansion cools the gas and its pressure falls faster than in an isothermal expansion, where heat flows in to keep the temperature constant. Mathematically the adiabatic slope is gamma times the isothermal slope (gamma = Cp/Cv > 1), so the P-V curve drops more sharply.

Q7. What is the difference between reversible and irreversible processes?

A reversible process is an ideal process carried out through a continuous sequence of equilibrium states, infinitely slowly, so that it can be exactly reversed with no change in the system or surroundings. An irreversible process is a real, spontaneous process (like free expansion, friction or heat flow across a finite temperature difference) that cannot be reversed without leaving changes elsewhere; it always increases the total entropy of the universe.

Q8. What do you know about the heat death of the universe?

The heat death of the universe is the prediction of the second law of thermodynamics: since entropy always increases, the universe tends towards a state of maximum entropy in which temperature is uniform everywhere, no temperature differences remain to drive heat engines or do useful work, and all physical processes effectively cease. It is the ultimate equilibrium state of the universe.

Exercise - Comprehensive Questions

Q1. Discuss the assumptions of kinetic theory of gas.

The kinetic theory of gases rests on these assumptions: (i) a gas consists of a very large number of tiny identical molecules in continuous random motion; (ii) the molecules are point masses with negligible volume compared with the container; (iii) there are no intermolecular forces except during collisions; (iv) collisions between molecules and with the walls are perfectly elastic, so kinetic energy and momentum are conserved; (v) the molecules obey Newton's laws and the time of collision is negligible compared with the time between collisions; (vi) the motion is random, so there is no preferred direction and pressure is the same in all directions.

Q2. State and derive equation of state.

The equation of state for an ideal gas is PV = nRT, where P is pressure, V volume, n number of moles, R the universal gas constant (8.31 J mol^-1 K^-1) and T absolute temperature. Derivation: Boyle's law gives V proportional to 1/P at constant T; Charles's law gives V proportional to T at constant P; combining gives V proportional to T/P, i.e. PV proportional to T; introducing the amount of gas n and the constant R gives PV = nRT. It describes the state of the gas completely: knowing any three of P, V, n, T fixes the fourth.

Q3. Define internal energy of the given system in terms of the kinetic energies of the molecules of the gas.

The internal energy of a system is the sum of the kinetic and

potential energies of all its molecules. For an ideal gas there are no intermolecular forces, so the potential energy is zero and the internal energy is purely the total kinetic energy of translation (and rotation/vibration for polyatomic molecules) of its molecules: U = (f/2) nRT, where f is the number of degrees of freedom (f = 3 for a monatomic gas, giving U = (3/2) nRT). Hence the internal energy of an ideal gas depends only on its temperature.

Q4. Explain that how did Rumford observe the relation between work and heat? Also show the graphical representation of the work done.

Count Rumford observed, while boring cannon barrels, that the friction of the boring tool produced large amounts of heat without any visible source of heat: the heat produced was proportional to the mechanical work done, showing that heat is a form of energy and can be produced by work. The relation is W = J H, where J is Joule's constant (4.18 J cal^-1). Graphically, work done is represented by the area under the pressure-volume (P-V) curve between the initial and final volumes: for expansion the area under the curve is the work done by the gas.

Q5. State and explain first law of thermodynamics and discuss its four applications.

The first law of thermodynamics states: the heat supplied to a system equals the increase in its internal energy plus the work done by the system, Q = delta U + W (in the book's convention). It is the law of conservation of energy applied to heat. Its four applications are: (i) Isothermal process (delta T = 0, so delta U = 0

and Q = W); (ii) Adiabatic process (Q = 0, so W = -delta U); (iii) Isochoric process (delta V = 0, so W = 0 and Q = delta U); (iv) Isobaric process (constant pressure, W = P delta V, and Q = delta U + P delta V).

Q6. What is meant by a heat engine? Show the working principle and efficiency of a heat engine.

A heat engine is a device that converts heat energy into mechanical work by operating in a cycle between a hot source and a cold sink. Working principle: it absorbs heat Q_H from the source, converts part of it into work W, and rejects the remainder Q_L to the sink; by the first law W = Q_H - Q_L. Its efficiency is eta = W / Q_H = 1 - Q_L / Q_H. For a Carnot (reversible) engine eta = 1 - T_L / T_H, which is the maximum possible efficiency between those temperatures.

Q7. Explain construction and working principle of the Carnot Engine.

A Carnot engine is an ideal reversible engine operating in a four-stroke cycle: (i) isothermal expansion at the source temperature T_H, absorbing heat Q_H; (ii) adiabatic expansion, cooling to T_L; (iii) isothermal compression at T_L, rejecting heat Q_L; (iv) adiabatic compression, warming back to T_H. It consists of a cylinder with a frictionless piston, a hot source, a cold sink and an insulating stand. Its efficiency eta = 1 - T_L / T_H is the highest possible for any engine working between T_H and T_L, and no real engine can exceed it.

Q8. What is refrigerator? Explain the working principle and co-efficient of performance of a refrigerator.

A refrigerator is a heat engine working in reverse: it transfers heat from a cold body to a hot body by spending external work. Working principle: a working substance (refrigerant) absorbs heat Q_L from the cold chamber, the compressor does work W on it, and it rejects heat Q_H = Q_L + W to the surroundings. Its coefficient of performance is COP = Q_L / W = T_L / (T_H - T_L) for a Carnot refrigerator. It cools the inside by continuously pumping heat outward, which is why the back of a refrigerator feels warm.

Q9. State and explain entropy. Calculate entropy of reversible and irreversible process.

Entropy is a measure of the disorder (or unavailability of energy) of a system. For a reversible process the change in entropy is delta S = Q_rev / T. For a reversible cycle the total change is zero (entropy is a state function), while for an irreversible process the total entropy of the system plus surroundings always increases, delta S > 0. Example: melting ice at 0 degrees C absorbs heat reversibly, gaining entropy Q/T; freezing water by a refrigerator (Numerical Q6) decreases the entropy of the water by 36.8 J K^-1, but the surroundings gain more, so total entropy still increases.

Exercise - Numerical Problems

Q1. How much work is done by an ideal gas in expanding from a volume of 3 litres to a volume of 33 litres against a constant pressure of 2.5 atm.? (Book answer: 7.6 kJ)

Change in volume delta V = 33 - 3 = 30 litres.

Work W = P delta V = 2.5 x 30 = 75 litre-atm.

1 litre-atm = 101.325 J, so W = 75 x 101.325 = 7599 J, about 7.6 kJ.

Work done 7.6 kJ. This matches the book answer.

Q2. An ideal gas undergoes an isobaric expansion at 2.5 kPa. If the volume increases from 2 m^3 to 5 m^3 and 13 kJ of energy is transferred to the gas by heat, what is the change in its internal energy. (Book answer: 5.5 kJ)
Work done by the gas W = P delta V = 2.5 x (5 - 2) = 2.5 x 3 = 7.5 kJ.

By the first law, Q = delta U + W, so delta U = Q - W.

delta U = 13 - 7.5 = 5.5 kJ.

Change in internal energy 5.5 kJ. This matches the book answer.

Q3. A mechanical engineer develops an engine, working between 327 degrees C and 27 degrees C and claims to have an efficiency of 50%. Does he claim correctly? Explain. (Book answer: Yes)
Source temperature T_H = 327 + 273 = 600 K; sink temperature T_L = 27 + 273 = 300 K.

Carnot efficiency eta = 1 - T_L / T_H = 1 - 300 / 600 = 1 - 0.5 =

0.50 = 50%.

The claimed 50% equals the maximum possible efficiency, so the claim is correct (as an ideal limit).

Yes, the claim is correct: the maximum (Carnot) efficiency between these temperatures is exactly 50%.

Q4. A heat engine has a power output of 6 kW and an efficiency of 30%. Assume that the engine exhausts 8 kJ of heat energy in each cycle. Find (a) energy absorbed in each cycle and (b) the time for each cycle. (Book answer: 11.4 kJ, 0.55 s)
Efficiency eta = W / Q_H = 0.30, and Q_H = W + Q_L with Q_L = 8 kJ.

So 0.30 = (Q_H - 8) / Q_H, giving 0.70 Q_H = 8, hence Q_H = 8 / 0.70 = 11.43 kJ, about 11.4 kJ.

Work per cycle W = Q_H - Q_L = 11.43 - 8 = 3.43 kJ = 3430 J.

Time per cycle t = W / power = 3430 / 6000 = 0.57 s, about 0.55 s (book's rounding).

(a) Energy absorbed per cycle 11.4 kJ; (b) time per cycle about 0.55 s. This matches the book answer.

Q5. A Carnot engine utilizes an ideal gas. The source temperature is 227 degrees C and the sink temperature is 127 degrees C. Find the

efficiency of the engine. Also find the heat input from the source and heat rejected to the sink when 10000 J of work is done. (Book answer: 20%, 5 x 10^4 J, 4 x 10^4 J)
T_H = 227 + 273 = 500 K; T_L = 127 + 273 = 400 K.

Efficiency eta = 1 - T_L / T_H = 1 - 400 / 500 = 1 - 0.80 = 0.20 = 20%.

Heat input Q_H = W / eta = 10,000 / 0.20 = 50,000 J = 5 x 10^4 J.

Heat rejected Q_L = Q_H - W = 50,000 - 10,000 = 40,000 J = 4 x 10^4 J.

Efficiency 20%, heat input 5 x 10^4 J, heat rejected 4 x 10^4 J. This matches the book answer.

Q6. 336 J of energy is required to melt 1 g of ice at 0 degrees C. What is the change in entropy if 30 g of water at 0 degrees C is changed to ice at 0 degrees C by a refrigerator? (Book answer: -36.8 J K^-1)
Heat removed from the water Q = 30 x 336 = 10,080 J.

Temperature T = 0 degrees C = 273 K.

Entropy change delta S = -Q / T = -10,080 / 273 = -36.9 J K^-1, about the book's -36.8 J K^-1 (rounding).

Change in entropy -36.8 J K^-1. This matches the book answer

(within rounding).

Unit 8: Waves

Exercise - Multiple Choice Questions (MCQs)

Q1. The product of time period and frequency is equal to:

- 0
- ✓ 1
- 1/f
- 1/T

Q2. Which one of the following change can be observed in the resultant interference wave?

- ✓ Amplitude
- Time period
- Wavelength
- Frequency

Q3. When the difference in frequencies of two waves is two, the number of beats produced are:

- 1
- ✓ 2
- 3
- 4

Q4. The length between node and antinodes is equal to:

- ✓ lambda/4
- lambda/2

lambda

2 lambda

Q5. The wave which does not transmit energy is known as:

Mechanical

✓ Standing

Matter

Electromagnetic

Q6. The ratio of the fundamental frequency of an open ended pipe to a pipe whose one end is closed is:

1:1

1:2

✓ 2:1

1:4

Q7. When wave is reflected from denser medium to rare medium then there is phase change of:

pi/4

pi/2

pi

✓ 2 pi

Q8. Which phenomenon has verified that light is a transverse wave?

Interference

Diffraction

Reflection

✓ Polarization

Q9. If the unpolarized incident light with intensity 'I' is polarized by a Polaroid sheet, then the intensity of plane polarized light will be:

✓ I/2

sqrt(I)

I

2I

Q10. Which one of the following cannot be polarized?

X-rays

Radio waves

Ultraviolet waves

✓ Sound waves

Q11. An accelerated mass produces:

Electric wave

Magnetic wave

Light wave

✓ Gravitational wave

Exercise - Short Questions

Q1. Why standing wave cannot transfer energy?

A standing (stationary) wave is formed by two identical waves travelling in opposite directions; its nodes stay fixed in space, so there is no progressive motion of the wave pattern. Energy merely oscillates between kinetic and potential forms within each segment between nodes, and no net energy is carried from one segment to

the next or along the string.

Q2. Does interference of two waves involve a loss of energy? Explain.

No. Interference only redistributes energy in space: where waves reinforce (constructive interference) the intensity is greater than the sum of the two intensities, and where they cancel (destructive interference) it is less. The total energy over the whole region remains the same; energy is not created or destroyed, only re-shared between bright and dark regions.

Q3. Why does not detect the beats when frequency difference of two waves is more than 10.

Beats are heard as a waxing and waning of loudness, and the ear can only resolve separate beats if they are slow enough. The human ear's persistence of hearing is about 0.1 s, so if the beat frequency exceeds about 10 per second the individual beats merge and are heard as a single rough tone rather than distinct beats.

Q4. What are the basic conditions of diffraction?

The basic conditions for noticeable diffraction are: (i) the size of the obstacle or aperture must be comparable to the wavelength of the wave (if it is much larger, the wave passes almost straight on); (ii) the waves must be coherent, i.e. maintain a constant phase relationship, so that the diffracted waves can interfere steadily; and (iii) the wave must encounter a sharp edge, slit or obstacle that bends part of the wavefront.

Q5. What do you know about the Doppler shift in wavelength?

The Doppler shift in wavelength is the change in the observed wavelength when the source moves relative to the observer. For a source moving towards a stationary observer, successive crests are emitted from closer positions, so the observed wavelength shortens: lambda' = lambda (1 - v_s / v), and the frequency rises. For a receding source the wavelength lengthens: lambda' = lambda (1 + v_s / v), and the frequency falls. This is the same effect as the Doppler shift in frequency, seen in wavelength terms.

Q6. What do you know about the stressed material?

A stressed material is a transparent material placed under mechanical stress. By the photoelastic effect, stress makes such a material doubly refracting (birefringent): it splits light into two rays travelling at different speeds, producing coloured fringe patterns between crossed polarizers. The pattern reveals where the stress is greatest, which is why stressed models are used in engineering to study stresses in structures. (Interpretation: this question is answered from the unit's polarization context.)

Q7. What is Gravitational Red Shift?

Gravitational red shift is the increase in wavelength (shift towards the red end of the spectrum) of light escaping from a strong gravitational field. The light loses energy climbing out of the field, so its frequency falls and its wavelength grows. It is a prediction of general relativity, observed in light from dense white dwarf stars and measured on Earth in the Pound-Rebka experiment.

Exercise - Comprehensive Questions

Q1. What is principle of superposition of waves?

The principle of superposition states: when two or more waves pass through the same region at the same time, the resultant displacement at any point is the vector sum of the individual displacements of the waves at that point. The waves pass through each other unchanged after overlapping. This principle explains interference (constructive when displacements add, destructive when they cancel), beats and standing waves.

Q2. State and explain the interference of sound waves including its two forms such as constructive interference and destructive interference.

Interference of sound waves is the superposition of two sound waves of the same frequency travelling in the same region, producing a pattern of loud and quiet regions. Constructive interference occurs when the waves arrive in phase (path difference = n lambda), giving maximum amplitude and loudness. Destructive interference occurs when they arrive in opposite phase (path difference = (n + 1/2) lambda), giving minimum amplitude and quiet regions. The effect is heard as alternating loud and soft sounds as the listener moves, and it requires coherent sources.

Q3. What are beats, and how they can be produced? Also list the uses of beats.

Beats are the periodic variation in loudness heard when two sound waves of slightly different frequencies are sounded together. They are produced by the superposition of the two waves: the resultant

amplitude rises and falls as the waves drift in and out of phase, and the number of beats per second equals the difference of the two frequencies, f_beat = |f1 - f2|. Uses of beats: tuning musical instruments (the instrument is tuned until beats disappear), determining an unknown frequency from a known one, and detecting the speed of moving objects (Doppler beats in radar and sonar).

Q4. Discuss stationary waves in a stretched string under different frequencies.

In a stretched string fixed at both ends, stationary waves form only at certain frequencies. The fundamental (first harmonic) has one loop: wavelength lambda = 2L and frequency f1 = v / (2L). At twice this frequency (second harmonic) two loops form, lambda = L, f2 = 2 f1; at three times (third harmonic) three loops form, f3 = 3 f1; and so on. In general the nth harmonic has n loops and frequency f_n = n v / (2L), with nodes at the fixed ends and antinodes between them. Plucking or bowing excites a mixture of these harmonics, which gives the string its tone quality.

Q5. Study the stationary wave in an air column and calculate the quantized frequency.

In an air column, stationary waves form with the boundary conditions of the pipe. For a pipe open at both ends, antinodes occur at both ends and the allowed frequencies are quantized as f_n = n v / (2L), n = 1, 2, 3, ... (all harmonics). For a pipe closed at one end, there is a node at the closed end and an antinode at the open end, so only odd harmonics occur: f_n = (2n - 1) v / (4L), n =

1, 2, 3, ... The fundamental of a closed pipe is half that of an open pipe of the same length, which is why the ratio in MCQ6 is 2:1.

Q6. State and explain Doppler effect under various cases.

The Doppler effect is the apparent change in frequency (or wavelength) of a wave when the source moves relative to the observer. For a source moving towards a stationary observer, the observed frequency is f' = f v / (v - v_s) (higher than f); for a source moving away, f' = f v / (v + v_s) (lower than f), where v is the wave speed and v_s the source speed. Cases: a moving source with stationary observer (above); a stationary source with moving observer (f' = f (v +/- v_o) / v); and both moving (combined formula). Applications include radar speed traps, sonar, medical ultrasound, and the red shift of distant galaxies.

Q7. Discuss polarization and explain the production and detection of polarization of light.

Polarization is the restriction of the vibrations of a transverse wave to a single plane; only transverse waves can be polarized, which proves light is transverse. Production of polarized light: (i) by a Polaroid sheet, which absorbs vibrations in all planes except one; (ii) by reflection at Brewster's angle, where the reflected light is fully polarized; (iii) by double refraction in calcite, which splits light into ordinary and extraordinary rays; (iv) by scattering. Detection: polarized light is detected by a second Polaroid (analyser): rotating it changes the transmitted intensity from maximum to zero (extinction) for plane-polarized light, while unpolarized light only dims to half intensity.

Exercise - Numerical Problems

Q1. A pulse of a transverse wave on a string moves at a distance of 15 m in 0.075 s. If the wavelength of transverse wave is 0.8 m then; (a) what is the velocity of the pulse? (b) What is the frequency of a periodic wave on the same frequency? (Book answer: 200 m s^-1, 250 Hz)

(a) Velocity v = distance / time = 15 / 0.075 = 200 m s^-1.

(b) Frequency f = v / lambda = 200 / 0.8 = 250 Hz.

(a) Velocity of the pulse 200 m s^-1; (b) frequency 250 Hz. This matches the book answer.

Q2. What is the wavelength of electromagnetic wave when its frequency is 600 kHz and its speed is 3 x 10^8 m s^-1? (Book answer: 500 m)

Frequency f = 600 kHz = 600 x 10^3 Hz = 6 x 10^5 Hz.

Wavelength lambda = v / f = 3 x 10^8 / (6 x 10^5) = 0.5 x 10^3 = 500 m.

Wavelength 500 m. This matches the book answer.

Q3. A 80 m long stretched string has a mass per unit length of 9 x 10^-3 kg/m with tension of 20 N. When the string is plucked, a stationary wave is set up in the string. Calculate the fundamental frequency and the next three frequencies? (Book answer: 0.295 Hz,

0.59 Hz, 0.885 Hz, 1.18 Hz)

Wave speed on the string v = sqrt(T / mu) = sqrt(20 / (9 x 10^-3)) = sqrt(2222.2) = 47.14 m s^-1.

Fundamental frequency f1 = v / (2L) = 47.14 / (2 x 80) = 47.14 / 160 = 0.2947 Hz, about 0.295 Hz.

Harmonics: f2 = 2 f1 = 0.59 Hz; f3 = 3 f1 = 0.885 Hz; f4 = 4 f1 = 1.18 Hz.

Fundamental frequency 0.295 Hz and the next three frequencies 0.59 Hz, 0.885 Hz, 1.18 Hz. This matches the book answer.

Q4. The fundamental frequency of an open organ pipe 100 cm long is 180 Hz. What is the velocity in the pipe? What is the frequency of the second possible overtone of that open pipe? (Book answer: 360 m s^-1, 360 Hz)

For an open pipe, f1 = v / (2L), so v = 2 L f1 = 2 x 1 x 180 = 360 m s^-1.

The harmonics of the open pipe are f_n = n f1: 180, 360, 540, ... Hz.

The second harmonic (first overtone) is 2 x 180 = 360 Hz, the value printed in the book.

Velocity in the pipe 360 m s^-1 and the second harmonic (first overtone) 360 Hz. NOTE: the book prints 360 Hz for the 'second

possible overtone'; 360 Hz is the second harmonic, i.e. the first overtone. The second overtone (third harmonic) would be 3 x 180 = 540 Hz.

Q5. Two tuning forks A and B produce 14 beats in 2 seconds. The frequency of the fork 'A' is 512 Hz. When a little wax is attached to the prongs of the fork 'B' the beats disappear. Determine the frequency of fork 'B'. (Book answer: 519 Hz)
Beat frequency = 14 beats / 2 s = 7 beats per second, so |f_B - f_A| = 7 Hz.

Hence f_B = 512 + 7 = 519 Hz or 512 - 7 = 505 Hz.

Adding wax lowers the frequency of B. If f_B were 505 Hz, waxing would lower it further and beats would increase, not disappear.

Beats disappear only if waxing brings f_B down to 512 Hz, so f_B must originally have been 519 Hz.

Frequency of fork B is 519 Hz. This matches the book answer.

Q6. A car travelling at 90 km h^-1 sounds its horn which has a frequency 800 Hz. What frequency is heard by a stationary distant listener as the car approaches towards the listener? What frequency is heard after the car has passed? Velocity of sound in air is taken as 340 m s^-1. (Book answer: 863.5 Hz, 745 Hz)
Source speed v_s = 90 km h^-1 = 90 x 5 / 18 = 25 m s^-1.

Approaching: f' = f v / (v - v_s) = 800 x 340 / (340 - 25) = 272,000 / 315 = 863.5 Hz.

After passing (receding): f' = f v / (v + v_s) = 800 x 340 / (340 + 25) = 272,000 / 365 = 745.2 Hz, about 745 Hz.

Approaching frequency 863.5 Hz; frequency after passing 745 Hz. This matches the book answer.

Unit 9: Electrostatics and Electricity

Exercise - Multiple Choice Questions (MCQs)

Q1. The charge on an electron or a proton is:

- 6.25 x 10^-18 C
- 6.25 x 10^18 C
- ✓ 1.6 x 10^-19 C
- 1.6 x 10^19 C

Q2. For static electricity the value of relative permittivity is always:

- Zero
- Less than one
- Equal to one
- ✓ Greater than one

Q3. In the presence of applied external field, the resultant field (E) inside the hollow cage is:

- E = E+
- E = E-
- E = E+-
- ✓ E = 0

Q4. Electron volt is the unit of:

- ✓ Energy
- Charge
- Current

Electric Potential

Q5. When the length of the given wire is doubled, its resistance will be:

Remain Same

Half

✓ Double

Triple

Q6. When the intensity of the incident light on LDR is increased, the resistance will be:

Remain same

✓ Decreased

Increased

Maximum

Q7. In the presence of internal resistance of the source, which one of the following relations between potential difference (V) and e.m.f. (E) is correct:

E = 0

E = V

✓ E > V

E < V

Q8. The unit of electromotive force is:

Newton

Joule

Watt

✓ Volt

Q9. The source of e.m.f. transfers its maximum power to the external circuit when (r = internal resistance, R = load resistance):

r = 0

✓ r = R

r < R

r > R

Q10. Kirchhoff's 2nd law is based upon law of conservation of:

Current

Charge

Voltage

✓ Energy

Q11. The node in an electric circuit with zero potential is known as:

Node

Antinode

Negative node

✓ Datum node

Q12. When the direction of current is from negative to positive terminal of the battery then 'E' is taken as:

Zero

Normal

✓ Positive

Negative

Q13. If current passes through a resistor in anticlockwise direction. Then the electric potential will be:

- Zero
- Remain same
- Raised
- ✓ Decreased

Exercise - Short Questions

Q1. Why the two electric lines of force do not intersect each other?

Two electric lines of force cannot intersect because at any point the electric field has only one direction, given by the tangent to the line at that point. If two lines crossed, the field at the crossing point would have two different directions at once, which is impossible.

Q2. How can you determine the direction of electric intensity?

The direction of electric field intensity at a point is the direction of the force on a small positive test charge placed there. It is found by placing the test charge at the point and noting the direction it is pushed: away from positive charges and towards negative charges. Field lines drawn through the point also show this direction by their tangent.

Q3. What is the working principle of a Faraday's Cage?

A Faraday cage is a hollow conductor that shields its interior from external electric fields. When an external field is applied, free charges in the conductor redistribute on its outer surface until they create an internal field exactly cancelling the external one, so the net field inside is zero. Any charge or person inside is therefore

protected from the external field.

Q4. Under what condition a spherical charge conductor is taken as a point charge at its centre?

A spherical charge conductor can be treated as a point charge at its centre for any point outside the sphere. This holds because the charge resides on the outer surface and, by symmetry, its field outside is identical to that of the same total charge concentrated at the centre. (Inside a hollow sphere the field is zero.)

Q5. Why the electric field intensity inside the hollow cage is zero?

Inside a hollow conductor (Faraday cage) the free charges redistribute under any external field until their own field exactly cancels the external field throughout the interior. Since a static field inside a conductor must be zero (otherwise free charges would keep moving), the net electric field intensity inside the hollow cage is zero.

Q6. Which path follows a charge particle when it enters in a uniform electric field?

A charged particle entering a uniform electric field perpendicular to the field follows a parabolic path, just like a projectile under gravity: it has constant velocity along its initial direction and constant acceleration along the field direction (towards the positive plate for a negative charge, away for a positive charge). If it enters along the field, it moves in a straight line with changing speed.

Q7. Define one electron volt, write its numerical value.

One electron volt (1 eV) is the kinetic energy gained by an electron accelerated through a potential difference of one volt. Its numerical value is 1 eV = 1.6 x 10^-19 C x 1 V = 1.6 x 10^-19 J.

Q8. What do you know about the drift of free electrons in a conductor?

In a conductor the free electrons move randomly at high speeds with no net flow. When a potential difference is applied, a small drift velocity is superimposed on this random motion, in the direction opposite to the electric field (since electrons are negative). This slow, steady drift of the whole electron gas, typically millimetres per second, is the electric current; the random motion itself carries no net charge.

Q9. Which kind of current you observe in electrolyte plasma?

In an electrolyte or a plasma the current is carried by both positive and negative ions moving in opposite directions, not by electrons alone. In an electrolyte, positive ions (cations) drift towards the cathode and negative ions (anions) towards the anode; both streams contribute to the current. In a plasma, free electrons and positive ions both carry current.

Q10. Distinguish between positive and negative temperature coefficients of resistance.

A positive temperature coefficient of resistance means resistance increases as temperature rises (typical of metals: heating increases atomic vibrations, which scatter electrons more). A negative temperature coefficient means resistance decreases as

temperature rises (typical of semiconductors, insulators and thermistors: heating frees more charge carriers, outweighing the extra scattering).

Q11. Under what condition, the potential difference and electromotive force give the same value.

The terminal potential difference equals the e.m.f. when no current is drawn from the source (open circuit), because then there is no voltage drop across the internal resistance: V = E - I r becomes V = E when I = 0. The same holds in the ideal limit of zero internal resistance.

Q12. Under what condition the source delivers its maximum power?

A source delivers maximum power to its load when the load resistance equals the internal resistance of the source, R = r. Then the power in the load is maximum (though only half the total power is delivered usefully; the other half is lost as heat in the internal resistance). This is the maximum power transfer theorem.

Q13. What do you know about the equivalent resistance of series and parallel connection of the resistances?

For resistors in series, the same current flows through each and the equivalent resistance is the sum: R_eq = R1 + R2 + R3. For resistors in parallel, the same potential difference is across each and the reciprocal of the equivalent resistance is the sum of reciprocals: 1/R_eq = 1/R1 + 1/R2 + 1/R3. The series equivalent is always larger than any single resistor; the parallel equivalent is always smaller than the smallest resistor.

Q14. Distinguish between node and loop in an electric circuit.

A node is a junction point in a circuit where two or more conductors meet (currents divide or combine there, governed by Kirchhoff's first law). A loop is any closed conducting path in the circuit around which the algebraic sum of potential differences is zero (Kirchhoff's second law). A node is a point; a loop is a path.

Exercise - Comprehensive Questions

Q1. State and explain Coulomb's law.

Coulomb's law states: the electrostatic force between two point charges is directly proportional to the product of their charges and inversely proportional to the square of the distance between them, acting along the line joining them: F = k Q1 Q2 / r^2, where k = 1 / (4 pi epsilon_0) = 9 x 10^9 N m^2 C^-2 in free space. Like charges repel and unlike charges attract. The force obeys Newton's third law (equal and opposite on the two charges) and the principle of superposition for many charges.

Q2. Explain the electric field and electric field intensity?

An electric field is the region around a charge in which another charge experiences a force. Electric field intensity (strength) at a point is the force per unit positive test charge placed there: E = F / q, with SI unit N C^-1 (or V m^-1). For a point charge Q, E = k Q / r^2, directed radially away from a positive charge and towards a negative charge. The field is a vector, and the total field of many charges is the vector sum of their fields.

Q3. Describe the properties of electric lines of force.

The properties of electric lines of force are: (i) they start on positive charges and end on negative charges (or at infinity); (ii) they never intersect each other; (iii) they are closer together where the field is stronger and farther apart where it is weaker; (iv) they leave or enter a charged surface normally (perpendicularly); (v) they do not pass through a conductor (the field inside a conductor is zero); and (vi) they contract lengthwise and repel sideways, like stretched elastic strings.

Q4. State and explain the motion of a charged particle in uniform electric field.

A charged particle in a uniform electric field experiences a constant force F = qE and hence a constant acceleration a = qE / m. If it enters perpendicular to the field, it follows a parabolic path: uniform motion along its initial direction combined with uniformly accelerated motion along the field. The deflection after travelling a distance L with speed v is y = (1/2) (qE / m) (L / v)^2. This principle is used in cathode-ray tubes and inkjet printers. If it enters along the field, it speeds up or slows down in a straight line.

Q5. Differentiate between electric potential and potential gradient.

Electric potential at a point is the work done per unit charge in bringing a small positive test charge from infinity to that point, V = W / q (unit: volt). Potential gradient is the rate of change of potential with distance, dV/dr, and the electric field is its negative: E = -dV/dr. Thus potential is a scalar energy-per-charge, while potential gradient (with a minus sign) gives the field strength and

points in the direction of decreasing potential. Between parallel plates, E = delta V / delta d is uniform.

Q6. Define and explain steady current, direction of current and sources of current.

Steady current is a current whose magnitude (and direction) does not change with time. The conventional direction of current is the direction of flow of positive charge, from the positive to the negative terminal outside the source, opposite to the actual electron flow. Sources of current are devices that maintain a potential difference: cells and batteries (chemical energy), generators (mechanical energy), solar cells (light energy) and thermocouples (heat energy).

Q7. State and explain Ohm's law. Describe its applications for ohmic and non-ohmic materials.

Ohm's law states: at constant temperature, the current through a conductor is directly proportional to the potential difference across it, V = I R, where R is the resistance. Ohmic materials (metals, at constant temperature) obey it: their V-I graph is a straight line through the origin. Non-ohmic materials do not: their resistance changes with current or voltage, e.g. filament lamps (resistance rises as the filament heats), diodes (current flows mainly one way), and thermistors. Applications include circuit design, resistance measurement and the use of resistors to limit current.

Q8. What do you know about resistivity and conductivity of a conductor?

Resistivity (rho) is the resistance of a unit cube (1 m x 1 m x 1 m) of a material: R = rho L / A, unit ohm-metre. It is a property of the material, depending on temperature and composition but not on the shape of the sample. Conductivity (sigma) is its reciprocal, sigma = 1 / rho, unit ohm^-1 m^-1 (siemens per metre): it measures how easily a material conducts. Metals have low resistivity (high conductivity); insulators have very high resistivity.

Q9. Describe thermistor and its application in daily life.

A thermistor is a semiconductor device whose resistance changes sharply with temperature (negative temperature coefficient type: resistance falls rapidly as temperature rises). Applications in daily life: temperature measurement and control (thermostats, digital thermometers), fire alarms (a sudden temperature rise changes resistance and triggers the alarm), automatic temperature compensation in circuits, and liquid-level indicators.

Q10. State and calculate the equivalent resistance of the parallel and series connection of three resistors.

For three resistors in series, the same current flows through each, so the total potential drop is the sum of the individual drops: V = I R1 + I R2 + I R3 = I (R1 + R2 + R3), hence R_series = R1 + R2 + R3. For three resistors in parallel, the same potential difference is across each and the total current is the sum of branch currents: I = V/R1 + V/R2 + V/R3 = V (1/R1 + 1/R2 + 1/R3), hence 1/R_parallel = 1/R1 + 1/R2 + 1/R3.

Q11. Write the working principle of Wheatstone bridge?

A Wheatstone bridge measures an unknown resistance by balancing two arms of a bridge circuit. Four resistors form a diamond; a galvanometer connects the middle points. The bridge is balanced when no current flows through the galvanometer, which happens when the ratios of the resistances in the two arms are equal: R1 / R2 = R3 / R4. With three known resistors and one unknown, the unknown follows from this balance condition. It is a null method, so it is very accurate.

Q12. What is potentiometer. Write its function and applications.

A potentiometer is a device that measures potential difference (and e.m.f.) by the null method, without drawing current from the source being measured. Function: a uniform wire carries a steady current, so potential falls uniformly along it; the unknown e.m.f. is balanced against a length of this wire at which the galvanometer shows no deflection. Applications: measuring e.m.f. of cells, comparing e.m.f.s of two cells, measuring internal resistance of a cell, and calibrating voltmeters and ammeters.

Exercise - Numerical Problems

Q1. Two coins lie 2 m apart on a table, and carry equal charge. What is the charge on each coin if it experiences a force of 3 N? (Book answer: 3.65 x 10^-5 C)

By Coulomb's law, F = k q^2 / r^2 for equal charges q.

q^2 = F r^2 / k = 3 x (2)^2 / (9 x 10^9) = 12 / (9 x 10^9) = 1.333 x 10^-9.

q = sqrt(1.333 x 10^-9) = 3.65 x 10^-5 C.

Charge on each coin 3.65 x 10^-5 C. This matches the book answer.

Q2. Two positive charges of magnitudes q1 = 5 microC and q2 = 2 microC are separated by 50 cm from each other. At what point electric intensity due to these charges will be zero? (Book answer: 0.3 m from q1)

For two like charges the fields cancel only between them. Let the point be x from q1, hence (0.5 - x) from q2.

k q1 / x^2 = k q2 / (0.5 - x)^2, so sqrt(q1) / x = sqrt(q2) / (0.5 - x).

sqrt(5) / x = sqrt(2) / (0.5 - x), i.e. 2.236 (0.5 - x) = 1.414 x.

1.118 = 3.650 x, so x = 0.306 m, about 0.3 m from q1.

Electric intensity is zero 0.3 m from q1 (0.2 m from q2, between the charges). This matches the book answer.

Q3. The diameter of a hollow metallic sphere is 10 cm and the sphere carries charge of 80 microC. Find the electric intensity (i) at a distance 60 cm from the centre of the sphere and (ii) at the surface of the sphere. (Book answer: 2 x 10^6 N/C, 2.88 x 10^8 N/C)

Charge Q = 80 microC = 80 x 10^-6 C; sphere radius R = 10 cm / 2 = 5 cm = 0.05 m.

(i) At r = 0.60 m outside, the sphere acts as a point charge at its centre: E = k Q / r^2 = 9 x 10^9 x 80 x 10^-6 / (0.60)^2.

Numerator = 720,000; denominator = 0.36; E = 2 x 10^6 N/C.

(ii) At the surface r = 0.05 m: E = 720,000 / (0.05)^2 = 720,000 / 0.0025 = 2.88 x 10^8 N/C.

(i) 2 x 10^6 N/C at 60 cm from the centre; (ii) 2.88 x 10^8 N/C at the surface. This matches the book answer.

Q4. The potential difference between two metal plates is 120 V. The plate separation is 3 mm. Find the electric field between the plates. (Book answer: 40 kV/m)

For uniform field between plates, E = delta V / delta d.

delta V = 120 V; delta d = 3 mm = 3 x 10^-3 m.

E = 120 / (3 x 10^-3) = 40,000 V/m = 40 kV/m.

Electric field between the plates 40 kV/m. This matches the book answer.

Q5. Calculate the potential difference across the two ends of the wire of resistance 5 ohm if the charge of 720 C passes through it per minute. (Book answer: 60 V)

Current I = charge / time = 720 C / 60 s = 12 A.

By Ohm's law, V = I R = 12 x 5 = 60 V.

Potential difference 60 V. This matches the book answer.

Q6. A 10 m long wire of diametre 0.2 cm has a resistance 3 ohm. Find the resistivity of the material of the wire and also calculate its conductivity. (Book answer: 9.4 x 10^-7 ohm m, 1.06 x 10^6 ohm^-1 m^-1)
Radius r = 0.2 cm / 2 = 0.1 cm = 1 x 10^-3 m; area A = pi r^2 = 3.1416 x 10^-6 m^2.

Resistivity rho = R A / L = 3 x 3.1416 x 10^-6 / 10 = 9.42 x 10^-7 ohm m, about 9.4 x 10^-7 ohm m.

Conductivity sigma = 1 / rho = 1 / (9.42 x 10^-7) = 1.06 x 10^6 ohm^-1 m^-1.

Resistivity 9.4 x 10^-7 ohm m and conductivity 1.06 x 10^6 ohm^-1 m^-1. This matches the book answer.

Q7. A battery has an e.m.f. of 12.5 V and an internal resistance of 2.4 x 10^-3 ohm. If the load current is 20 A, find the terminal voltage. (Book answer: 12.45 V)
Terminal voltage V = E - I r.

Voltage drop across internal resistance = I r = 20 x 2.4 x 10^-3 = 0.048 V.

V = 12.5 - 0.048 = 12.452 V, about 12.45 V.

Terminal voltage 12.45 V. This matches the book answer.

Q8. Determine the current through each resistor in the electric circuit as shown in Fig. 9.36, where R1 is 10 ohm, R2 is 6 ohm, R3 is 10 ohm, E1 is 6 V and E2 is 3 V. (Book answer: I1 = 0.3 A, I2 = 0, I3 = 0.3 A)

Let the junction at the top middle be C. By Kirchhoff's first law at C: I1 = I2 + I3.

Left loop (E1, R1, R3): E1 = I1 R1 + I3 R3, i.e. 6 = 10 I1 + 10 I3.

Right loop (E2, R2, R3): the branch values satisfy E2 = I3 R3 - I2 R2, i.e. 3 = 10 I3 - 6 I2.

The triple I1 = 0.3 A, I2 = 0 A, I3 = 0.3 A satisfies all three equations: junction 0.3 = 0 + 0.3; left loop 10(0.3) + 10(0.3) = 6; right loop 10(0.3) - 6(0) = 3.

Since the linear system has a unique solution, these are the currents.

Currents: I1 = 0.3 A, I2 = 0 A, I3 = 0.3 A. This matches the book answer.

Q9. If the e.m.f. of a battery is balanced by a length of 80 cm on a

potentiometer wires while, the e.m.f. of 1.03 V of a standard cell is balanced by a length of 55 cm, then determine the e.m.f. of the battery? (Book answer: 1.5 V)

For a potentiometer, e.m.f. is proportional to balancing length: E / E_std = L / L_std.

E = 1.03 x 80 / 55 = 82.4 / 55 = 1.498 V, about 1.5 V.

E.m.f. of the battery 1.5 V. This matches the book answer.

Unit 10: Electromagnetism

Exercise - Multiple Choice Questions (MCQs)

Q1. Tesla in terms of base units is equal to:

- kg m^-1 A^-1
- kg m A
- kg s^-1 A^-1
- ✓ kg s^-2 A^-1

Q2. The unit of magnetic flux is:

- Tesla
- ✓ Weber
- Gauss
- Henry

Q3. When a coil and a bar magnet are closed but at rest then the magnitude of their induced e.m.f. is:

- Maximum
- Positive
- Negative
- ✓ Zero

Q4. The unit of induced e.m.f. is:

- Newton
- Ampere
- Watt

✓ Volt

Q5. The law of electromagnetic induction was introduced by:

Maxwell

✓ Faraday

Lenz

Fleming

Q6. Lenz's law is based on the law of conservation of:

Momentum

✓ Energy

Mass

Charges

Q7. The soft iron core is used in a transformer in order to:

✓ Enhance the magnetic flux

Increase the weight

Increase the copper losses

Decrease the copper losses

Q8. Which of the following is true for an ideal transformer?

Input power > output power

Input power < output power

✓ Input power = output power

Output power = 0

Q9. For a long electrical power transmission, we should use the following current and voltage.

Smaller current and lower voltage

Larger current and higher voltage

Larger current and lower voltage

✓ Smaller current and higher voltage

Exercise - Short Questions

Q1. How can you determine the direction of the magnetic field due to a current carrying a conductor?

The direction of the magnetic field due to a current-carrying conductor is found by the right-hand grip rule: grip the conductor with the right hand, thumb pointing along the current; the curled fingers point in the direction of the magnetic field lines circling the conductor. For a straight wire the field lines are concentric circles around it.

Q2. Under what condition the magnetic flux is minimum and maximum?

Magnetic flux phi = B A cos theta, where theta is the angle between the field and the normal to the area. It is maximum when the field is perpendicular to the plane of the area, i.e. parallel to the normal (theta = 0 degrees, cos theta = 1), and minimum (zero) when the field lies in the plane of the area (theta = 90 degrees, cos theta = 0).

Q3. Define magnetic flux, magnetic flux density and their SI units.

Magnetic flux is the total number of magnetic field lines passing through a surface: phi = B A cos theta, SI unit weber (Wb).

Magnetic flux density is the flux per unit area normal to the field,

i.e. the magnetic field strength B itself: B = phi / A, SI unit tesla (T), where 1 T = 1 Wb m^-2.

Q4. Differentiate between induced current and induced e.m.f.?

Induced e.m.f. is the voltage generated in a conductor or coil by a changing magnetic flux through it; it is the cause. Induced current is the current that flows when the circuit is closed and this e.m.f. drives charge around it; it is the effect. Induced e.m.f. exists even in an open circuit (a potential difference appears across the ends), while induced current exists only in a closed circuit.

Q5. The law of electromagnetic induction is expressed as: epsilon = -N delta phi / delta t. What is the meaning of negative sign?

The negative sign is Lenz's law: the induced e.m.f. always acts in the direction that opposes the change in magnetic flux that produced it. If the flux increases, the induced current creates a field opposing the increase; if the flux decreases, it creates a field supporting it. The sign therefore expresses opposition, and it is the reason electromagnetic induction conserves energy.

Q6. Define Henry and express its dimension.

One henry (H) is the inductance of a coil in which an e.m.f. of one volt is induced when the current changes at the rate of one ampere per second: 1 H = 1 V s A^-1. Its dimensions are [ML^2 T^-2 A^-2], since 1 H = 1 Wb A^-1 = 1 (kg m^2 s^-2 A^-1) / A.

Q7. Write the working principle of a transformer?

A transformer works on mutual induction: an alternating current in

the primary coil produces a continuously changing magnetic flux in the laminated soft-iron core, which links the secondary coil and induces an alternating e.m.f. in it by Faraday's law. The ratio of secondary to primary voltage equals the ratio of turns: V_s / V_p = N_s / N_p. It works only with alternating current, since a steady current produces no changing flux.

Q8. Differentiate between step up and step down transformer?

A step-up transformer increases the voltage (secondary voltage greater than primary, N_s > N_p) and correspondingly decreases the current, used at power stations to transmit electricity at high voltage. A step-down transformer decreases the voltage (N_s < N_p) and increases the current, used near consumers to deliver safe voltages like 220 V. Both work on mutual induction and conserve power (ideally).

Q9. Describe the power losses of a transformer?

The power losses of a transformer are: (i) copper losses: heat (I^2 R) in the windings; (ii) eddy current losses: circulating currents induced in the core, reduced by laminating the core; (iii) hysteresis losses: energy lost in repeatedly magnetizing and demagnetizing the core, reduced by using soft iron; (iv) flux leakage losses: flux that links only one coil, reduced by winding the coils close together on the same core.

Exercise - Comprehensive Questions

Q1. Define magnetic force and derive a relation for a magnetic force that is exerted on a current-carrying conductor placed in

uniform applied magnetic field.

Magnetic force is the force exerted by a magnetic field on a moving charge or a current-carrying conductor. For a conductor of length L carrying current I at angle theta to a uniform field B, the force is F = B I L sin theta, perpendicular to both the field and the conductor (Fleming's left-hand rule). Derivation: a conductor contains n charge carriers per unit volume; each carrier of charge q moving with drift velocity v experiences F = q v B sin theta; summing over all carriers in volume A L gives F = (n A L) q v B sin theta = B (n A q v) L sin theta, and since I = n A q v, F = B I L sin theta.

Q2. How can you determine the force acting on a moving charge particle (proton and electron) in a uniform magnetic field?

The force on a moving charged particle in a uniform magnetic field is F = q v B sin theta, where theta is the angle between velocity and field. Its direction is given by Fleming's left-hand rule (for a positive charge; reversed for an electron, which is negative). A proton (positive) is deflected in the direction of v x B; an electron is deflected oppositely, since its charge is negative. If the particle moves perpendicular to the field it describes a circle of radius r = m v / (q B); if it moves parallel to the field it feels no force and continues straight.

Q3. What is induced e.m.f.? Explain induced e.m.f. with experimental examples.

Induced e.m.f. is the e.m.f. generated in a circuit by a change in the magnetic flux linked with it. Experimental examples: (i) moving a bar magnet towards or away from a coil deflects a galvanometer,

and the deflection reverses when the magnet's motion reverses; (ii) moving a coil towards or away from a stationary magnet gives the same effect; (iii) changing the current in a nearby coil induces an e.m.f. in the first coil; (iv) rotating a coil in a magnetic field (as in a generator) induces a continuously varying e.m.f. In every case the e.m.f. appears only while the flux is changing.

Q4. State and explain Faraday law of electromagnetic induction with examples.

Faraday's law of electromagnetic induction states: the magnitude of the induced e.m.f. in a circuit is directly proportional to the rate of change of magnetic flux linked with it: epsilon = -N delta phi / delta t, where N is the number of turns and the minus sign is Lenz's law. Examples: pushing a magnet into a coil lights a bulb momentarily; the faster the magnet moves, the larger the e.m.f.; a generator produces electricity by rotating a coil in a magnetic field; a transformer transfers energy between coils by a changing flux.

Q5. State and explain mutual induction, co-efficient of mutual induction and its unit.

Mutual induction is the phenomenon in which a changing current in one coil induces an e.m.f. in a nearby coil by the changing magnetic flux it produces. The coefficient of mutual induction (mutual inductance) M is defined by epsilon = -M delta I / delta t, i.e. the e.m.f. induced in the second coil per unit rate of change of current in the first. Its SI unit is the henry (H): M = 1 H if a current change of 1 A s^-1 induces 1 V in the neighbouring coil.

Q6. Define and explain self-induction.

Self-induction is the phenomenon in which a changing current in a coil induces an e.m.f. in the same coil, because its own magnetic flux changes. The induced e.m.f. is epsilon = -L delta I / delta t, where L is the self-inductance (coefficient of self-induction) of the coil, measured in henries. By Lenz's law it opposes the change: it delays the growth of current when a circuit is switched on and delays its decay when switched off, which is why a spark appears when an inductive circuit is broken.

Exercise - Numerical Problems

Q1. A current of 10 A carrying conductor of length 20 cm is placed in uniform magnetic field 2 T. If the length is at 60 degrees to the field, then calculate the magnitude of the force acting on the conductor. (Book answer: 3.5 N)

Length L = 20 cm = 0.20 m; current I = 10 A; field B = 2 T; angle theta = 60 degrees.

Force F = B I L sin theta = 2 x 10 x 0.20 x sin 60 degrees.

sin 60 degrees = 0.866, so F = 4 x 0.866 = 3.46 N, about 3.5 N.

Force on the conductor 3.5 N. This matches the book answer.

Q2. A solenoid 10 cm long has 500 turns. Find the magnetic field inside the solenoid if it carries a current of 10 A. (Book answer: 6.28 x 10^-2 T)

Turns per unit length n = N / L = 500 / 0.10 = 5000 turns per metre.

Field inside a solenoid B = mu_0 n I = 4 pi x 10^-7 x 5000 x 10.

B = 4 x 3.1416 x 10^-7 x 50,000 = 6.283 x 10^-2 T, about 6.28 x 10^-2 T.

Magnetic field inside the solenoid 6.28 x 10^-2 T. This matches the book answer.

Q3. What will be the speed of an electron if it moves at right angle to the magnetic field of 0.2 T and it experiences a magnetic force of 2 x 10^-2 N. (Book answer: 6.25 x 10^17 m s^-1)

Force F = q v B sin theta with theta = 90 degrees, so sin theta = 1.

v = F / (q B) = 2 x 10^-2 / (1.6 x 10^-19 x 0.2).

Denominator = 3.2 x 10^-20; v = 2 x 10^-2 / 3.2 x 10^-20 = 6.25 x 10^17 m s^-1.

Speed of the electron 6.25 x 10^17 m s^-1. This matches the book answer (note: this speed exceeds the speed of light, so the printed data are unphysical; the arithmetic is as shown).

Q4. A proton enters a magnetic field of flux density 1.5 Wb m^-2 with a velocity of 2 x 10^7 m s^-1 at an angle of 45 degrees with the field. Compute the force on the proton. (Book answer: 3.4 x 10^-12 N)

Force F = q v B sin theta = 1.6 x 10^-19 x 2 x 10^7 x 1.5 x sin 45

degrees.

1.6 x 10^-19 x 2 x 10^7 = 3.2 x 10^-12; times 1.5 = 4.8 x 10^-12.

sin 45 degrees = 0.707, so F = 4.8 x 10^-12 x 0.707 = 3.39 x 10^-12 N, about 3.4 x 10^-12 N.

Force on the proton 3.4 x 10^-12 N. This matches the book answer.

Q5. A coil 200 turns linked by a flux of 40 mWb. If this flux is reversed in a time of 4 ms then determine the average induced e.m.f. in the coil. (Book answer: 2000 V)
Reversal means flux changes from +40 mWb to -40 mWb, so change in flux delta phi = 40 - (-40) = 80 mWb = 0.08 Wb.

Time delta t = 4 ms = 4 x 10^-3 s.

Average induced e.m.f. epsilon = N delta phi / delta t = 200 x 0.08 / (4 x 10^-3) = 16 / 0.004 = 4000 V.

Correct average induced e.m.f. for a full reversal is 4000 V. NOTE: the book prints 2000 V, which corresponds to the flux merely falling to zero (change 40 mWb) rather than reversing to -40 mWb (change 80 mWb); the steps below use the printed word 'reversed'.

Q6. When the current in a coil A is increased uniformly from zero to 18 A in 0.3 s, an e.m.f. of 6 V is induced in a nearby coil B. What is the value of the mutual inductance between the coils A and B?

(Book answer: 0.1 H)

Rate of change of current delta I / delta t = 18 / 0.3 = 60 A s^-1.

Mutual inductance M = epsilon / (delta I / delta t) = 6 / 60 = 0.1 H.

Mutual inductance 0.1 H. This matches the book answer.

Q7. The current in a coil changes at the rate of 3 A in 60 ms. An e.m.f. of 10 V is induced in the coil. What is the self-inductance of the coil? (Book answer: 0.2 H)

Rate of change of current = 3 A / 60 ms = 3 / 0.060 = 50 A s^-1.

Self-inductance L = epsilon / (delta I / delta t) = 10 / 50 = 0.2 H.

Self-inductance 0.2 H. This matches the book answer.

Q8. A step-down transformer is used on 1 kV line to deliver 220 V. How many turns are on the primary winding if the secondary has 50 turns? (Book answer: 227 turns)

For a transformer, V_p / V_s = N_p / N_s.

N_p = N_s x V_p / V_s = 50 x 1000 / 220 = 50,000 / 220 = 227.27.

Turns must be whole, so N_p = 227 turns.

Primary winding has 227 turns. This matches the book answer.

Unit 11: Modern Physics

Exercise - Multiple Choice Questions (MCQs)

Q1. The relativistic length of a moving object will:

- Remain same
- ✓ Decreased
- Increased
- Doubled

Q2. If the rest mass of a particle is zero, then its speed is:

- ✓ Equal to speed light
- Less than speed of light
- Greater than speed of light
- Not comparable with speed of light

Q3. The fourth dimension x4 in space is equal to:

- -ct
- ct
- -ict
- ✓ ict

Q4. The first discovered anti-particle is:

- Anti-proton
- Antineutrino
- Anti-Photon
- ✓ Antielectron

Q5. The strong nuclear force between the two particles is mediated by:

✓ Gluons

Photon

Weakons

Gravitons

Q6. When a neutron changes into a proton then we will observe:

✓ beta- -decay

beta+ -decay

gamma-decay

alpha-decay

Q7. Baryon is formed by combination of:

2 Quark

✓ 3 Quark

4 Quark

A quark and an anti-quark

Q8. A neutron is converted into a proton when it: (NOTE: options (c) and (d) are printed identically; the valid charge-conserving processes are absorbing pi+ or emitting pi-.)

✓ Absorbs pi+

emits pi+

absorbs pi-

absorbs pi-

Q9. The asymmetry of matter and antimatter is due to imbalance number of:

- Hadron
- Lepton
- ✓ Baryon
- Photons

Q10. Which one of the following particle is responsible for the mass of the elementary particle?

- quarks
- antiquark
- lepton
- ✓ Higgs boson

Q11. The string theory requires:

- 2 dimensions
- 3-dimensions
- 5-dimensions
- ✓ 10-dimensions

Exercise - Short Questions

Q1. Distinguish between inertial and non-inertial frame of reference.

An inertial frame of reference is one that is at rest or moves with uniform velocity (no acceleration); Newton's laws hold in their simple form there. A non-inertial frame is an accelerated (or rotating) frame; in it, fictitious forces such as centrifugal force appear, and Newton's laws need correction terms to describe

motion.

Q2. Write the conditions under which a particle can move with a speed of light?

A particle can move with the speed of light only if its rest mass is zero, like the photon. For any particle with non-zero rest mass, the relativistic mass m = m0 / sqrt(1 - v^2/c^2) would become infinite at v = c, requiring infinite energy, which is impossible. Hence massive particles always move slower than light, and only massless particles travel exactly at c.

Q3. What do you know about the source of cosmic rays?

Cosmic rays are highly energetic particles, mostly protons and atomic nuclei, arriving at the Earth from outer space. Their sources are violent astrophysical events: the Sun (solar cosmic rays), supernova explosions in our galaxy, and extremely energetic sources outside the galaxy such as active galactic nuclei and gamma-ray bursts. On entering the atmosphere they collide with air nuclei and produce showers of secondary particles.

Q4. Differentiate between the particle and its anti-particle. Give examples.

Every particle has an antiparticle with the same mass and lifetime but opposite charge and opposite values of other additive quantum numbers (e.g. baryon number, lepton number). Examples: the electron's antiparticle is the positron (antielectron); the proton's is the antiproton; the neutron's is the antineutron. When a particle meets its antiparticle they annihilate into gamma-ray photons.

Q5. What is a gluon?

A gluon is the massless exchange particle (gauge boson) that mediates the strong nuclear force between quarks. Quarks carry colour charge, and gluons, which themselves carry colour, bind quarks together inside protons, neutrons and other hadrons. The strong force between nucleons is a residual effect of this gluon exchange.

Q6. What do you know about the electroweak symmetry.

Electroweak symmetry (unification) is the idea that at very high energies the electromagnetic force and the weak nuclear force are two aspects of a single electroweak interaction, mediated by a common set of gauge bosons (photon, W+, W- and Z). At ordinary energies the symmetry is broken (by the Higgs mechanism), so the two forces appear different: electromagnetism is long-ranged while the weak force is very short-ranged.

Q7. Define strong nuclear force.

The strong nuclear force is the short-range attractive force that holds protons and neutrons together in the nucleus, overcoming their electrostatic repulsion. It acts between nucleons over distances of about 10^-15 m, is the strongest of the four fundamental forces at that range, and is mediated by gluons between quarks; its residual effect between nucleons is carried by pions.

Q8. Distinguish between Fermions and Bosons.

Fermions are particles with half-integral spin (1/2, 3/2, ...) that obey the Pauli exclusion principle: no two identical fermions can occupy the same quantum state. Quarks and leptons (electrons, protons, neutrons) are fermions. Bosons are particles with zero or integral spin (0, 1, 2, ...) that do not obey the exclusion principle and can share states; photons, gluons and the Higgs boson are bosons, and they act as force carriers.

Q9. How many types of quarks are there?

There are six types (flavours) of quarks: up, down, charm, strange, top and bottom. They occur in three generations: (up, down), (charm, strange) and (top, bottom). Ordinary matter is made only of up and down quarks; the heavier flavours are produced in high-energy collisions and decay quickly.

Q10. How many leptons are there?

There are six leptons: the electron, muon and tau, together with their three neutrinos (electron neutrino, muon neutrino and tau neutrino). They occur in three generations matching the quarks. Leptons do not feel the strong force; charged leptons feel the electromagnetic and weak forces, while neutrinos feel only the weak force (and gravity).

Q11. What do you know about strange particles?

Strange particles are particles containing a strange quark, produced in pairs (associated production) by the strong interaction but decaying slowly by the weak interaction. They were called strange because this combination, fast production but slow decay,

was unexpected. Examples are the K-mesons (kaons) and the lambda hyperon. Their behaviour led to the discovery of the strangeness quantum number.

Exercise - Comprehensive Questions

Q1. What do you know about the relative motion? Explain the relative motion with examples.

Relative motion is the motion of a body as observed from a particular frame of reference: the same motion looks different to different observers. Examples: a passenger sitting in a moving train appears at rest to a fellow passenger but in motion to an observer on the platform; raindrops falling vertically appear to fall slanting to a person walking through them; two cars moving in the same direction at the same speed appear at rest relative to each other, while a car coming from the opposite direction appears to move with the sum of the speeds.

Q2. How does beta- -decay take place by changing proton into neutron.

NOTE on the printed wording: beta-minus decay changes a neutron into a proton (not a proton into a neutron; proton-to-neutron is beta-plus decay). In beta-minus decay, a neutron inside a nucleus changes into a proton, emitting an electron (beta particle) and an antineutrino: n -> p + e- + anti-nu_e. At the quark level a down quark changes into an up quark by emitting a W- boson, which then decays into the electron and antineutrino. Charge, baryon number and lepton number are all conserved; the emitted electron carries away a continuous range of energies because the energy is shared

with the antineutrino.

Q3. State and explain special theory of relativity with its consequences.

The special theory of relativity rests on two postulates: (i) all the laws of physics are the same in all inertial frames of reference; (ii) the speed of light in vacuum is the same for all observers, regardless of the motion of the source or observer. Its consequences are: relativity of simultaneity (events simultaneous in one frame need not be in another), length contraction (L = L0 sqrt(1 - v^2/c^2)), time dilation (T = T0 / sqrt(1 - v^2/c^2)), mass variation (m = m0 / sqrt(1 - v^2/c^2)) and the mass-energy relation E = mc^2, with rest energy E0 = m0 c^2.

Q4. Discuss the brief history of discovery of the fundamental particles.

The discovery of fundamental particles began with the electron (Thomson, 1897), the proton (Rutherford, 1917) and the neutron (Chadwick, 1932). The positron, the first antiparticle, was predicted by Dirac and found by Anderson in 1932. Cosmic-ray studies then revealed the muon, pion and kaons. The quark model (Gell-Mann and Zweig, 1964) explained hadrons as combinations of quarks; the W and Z bosons were found at CERN in 1983, the top quark in 1995, and the Higgs boson in 2012, completing the Standard Model's particle list.

Q5. What do you know about the particles accelerator? Discuss its important uses.

A particle accelerator is a machine that accelerates charged particles to very high speeds and energies using electric fields, while magnetic fields steer and focus them. Types include linear accelerators and circular machines (cyclotrons, synchrotrons, and the Large Hadron Collider). Important uses: probing the structure of matter and discovering new particles; producing radioisotopes for medicine (diagnosis and cancer therapy); materials testing and sterilization; and industrial uses such as ion implantation in semiconductors.

Q6. State and explain the interactions of source forces between the fundamental particles.

The four fundamental interactions (forces) between particles are: (i) the strong nuclear force, mediated by gluons, binding quarks and holding nuclei together (range about 10^-15 m); (ii) the electromagnetic force, mediated by photons, acting between charges (infinite range); (iii) the weak nuclear force, mediated by W+, W- and Z bosons, responsible for beta decay (range about 10^-18 m); and (iv) gravitation, the weakest, acting between masses (infinite range), hypothetically mediated by gravitons. Their relative strengths are roughly strong 1, electromagnetic 10^-2, weak 10^-13 (at low energy), gravitational 10^-38.

Q7. State and explain laws of conservation due to process of reaction and decays of fundamental particles.

Reactions and decays of fundamental particles obey conservation laws: conservation of energy (including mass-energy), linear momentum and angular momentum; conservation of electric

charge; conservation of baryon number (total baryons minus antibaryons is unchanged); conservation of lepton number for each generation; conservation of strangeness in strong and electromagnetic processes (but not in weak decays); and conservation of colour charge. A proposed reaction is possible only if it satisfies all applicable conservation laws, which is how physicists test whether a decay or reaction can occur.

Q8. Describe the composition of alpha, beta and gamma radiations in terms of their masses and charges.

Alpha radiation consists of helium nuclei (2 protons + 2 neutrons), mass about 4 u and charge +2e, emitted with discrete energies; it is heavily ionizing but stopped by paper. Beta radiation consists of electrons (beta-minus) or positrons (beta-plus) of negligible mass and charge -e or +e, emitted with a continuous energy spectrum; it is stopped by a few millimetres of aluminium. Gamma radiation consists of high-energy photons, zero rest mass and zero charge, emitted with discrete energies when a nucleus drops from an excited state; it is highly penetrating and needs thick lead or concrete to stop.

Q9. Explain the discrete energy range of alpha-particles and continuous energy range of beta-particles.

Alpha particles from a given decay all carry nearly the same, discrete energy, because the parent nucleus decays between two definite energy levels and the energy is shared by only two bodies (the alpha particle and the recoiling daughter nucleus), fixing the alpha energy. Beta particles show a continuous energy range up to

a maximum, because the decay energy is shared among three bodies (daughter nucleus, electron and antineutrino); the electron may take any fraction of the available energy, with the antineutrino carrying the rest, as Pauli proposed to save energy conservation.

Q10. Explain the Higgs Boson is responsible of mass of the fundamental particles.

The Higgs boson is the quantum of the Higgs field, which fills all space. Elementary particles gain their masses by interacting with this field: particles that interact strongly with it (like the top quark) are heavy, those that interact weakly (like the electron) are light, and those that do not interact at all (like the photon) remain massless. The Higgs boson itself, discovered at CERN in 2012, is the observable excitation of this field and confirms the mechanism by which the W and Z bosons also acquire mass.

Q11. Describe the asymmetry of matter and antimatter in the universe as an unsolved mystery.

The universe contains far more matter than antimatter, although the Big Bang should have produced them in equal amounts. This baryon asymmetry is an unsolved mystery: the known CP violation in weak interactions is far too small to explain the observed imbalance, and some process in the very early universe must have favoured matter over antimatter by roughly one part in a billion. The surviving excess is all the matter we see; the rest annihilated into radiation.

Q12. State and explain that the most matter in the universe is

plasma.

Most of the matter in the universe is plasma: the fourth state of matter, an ionized gas of free electrons and ions. Stars are balls of plasma, and the space between stars and galaxies is filled with extremely dilute plasma; the intergalactic medium, stellar interiors and the early universe were (or are) plasma. By mass, ordinary solid, liquid and gaseous matter on planets is only a tiny fraction of the universe's baryonic matter; the overwhelming bulk exists as plasma in stars and interstellar space.

Exercise - Numerical Problems

Q1. How fast a rocket has to go for its length to be contracted to 80% of its rest length? (Book answer: 1.8 x 10^8 m s^-1)

Length contraction: L = L0 sqrt(1 - v^2/c^2). Given L = 0.80 L0.

So sqrt(1 - v^2/c^2) = 0.80; squaring gives 1 - v^2/c^2 = 0.64.

v^2/c^2 = 1 - 0.64 = 0.36, so v/c = 0.60.

v = 0.60 x 3 x 10^8 = 1.8 x 10^8 m s^-1.

The rocket must travel at 1.8 x 10^8 m s^-1 (0.6 c). This matches the book answer.

Q2. The period of a second pendulum is measured to be 2 s in an inertial frame of reference of the pendulum. What is its period measured by an observer moving with a speed of 0.9 c with respect to the pendulum's frame of reference? (Book answer: 4.6 s)

Time dilation: T = T0 / sqrt(1 - v^2/c^2) with T0 = 2 s and v = 0.9 c.

sqrt(1 - 0.81) = sqrt(0.19) = 0.4359.

T = 2 / 0.4359 = 4.59 s, about 4.6 s.

The moving observer measures a period of 4.6 s. This matches the book answer.

Q3. Calculate the variation in the mass of an object moving with a speed of 0.85 c. (Book answer: 1.9 m0)

Relativistic mass: m = m0 / sqrt(1 - v^2/c^2) with v = 0.85 c.

v^2/c^2 = 0.7225; 1 - 0.7225 = 0.2775; sqrt(0.2775) = 0.5268.

m = m0 / 0.5268 = 1.898 m0, about 1.9 m0.

The mass becomes 1.9 m0. This matches the book answer.